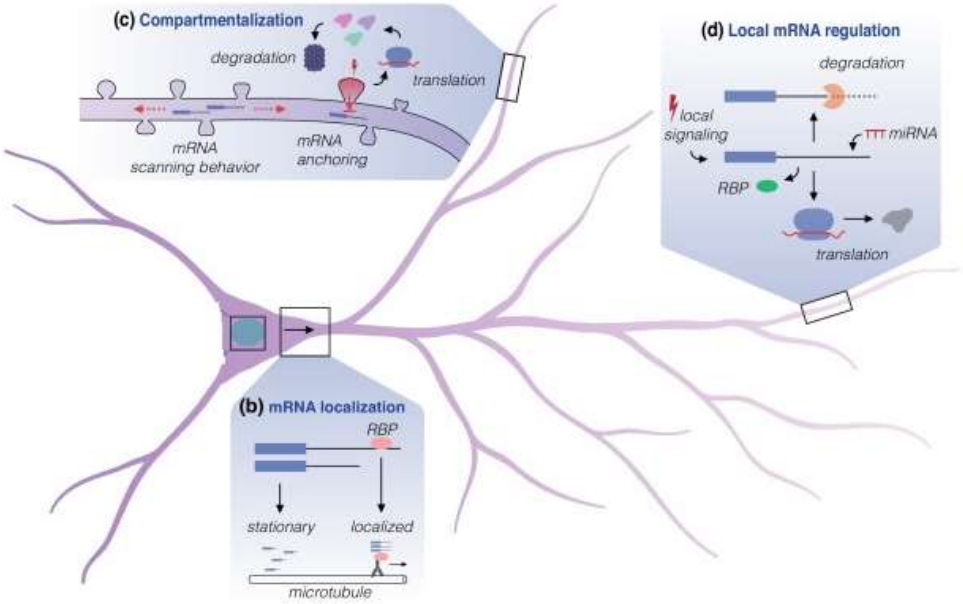
Spatial transcriptome profiling by MERFISH reveals sub-cellular RNA compartmentalization and cell-cycle dependent gene expression

Jean Fan, PhD
NCI F99/K00 Fellow
Zhuang Lab
Harvard University

Spatially-resolved transcriptomic characterization by imaging can reveal the spatial distribution and quantities of genes at subcellular and cellular levels

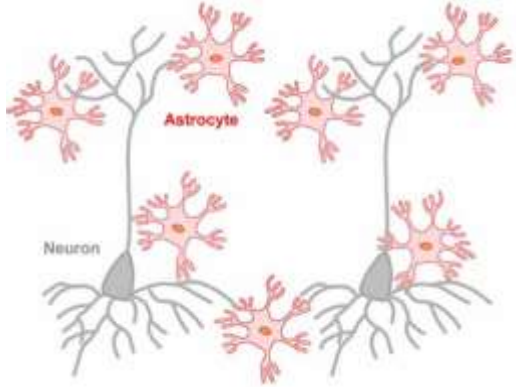
subcellular level



(Glock et al, Current Opinions in Neurobiology 2017)

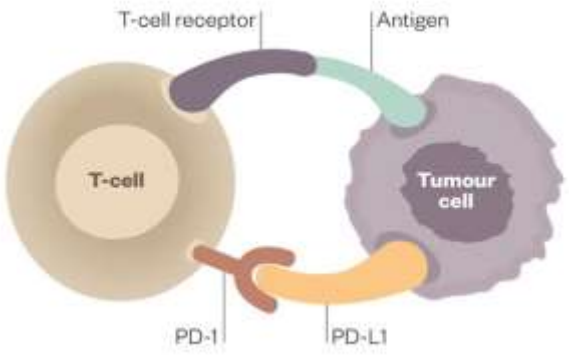
cellular/tissue level

Neuronal-glial interaction in development



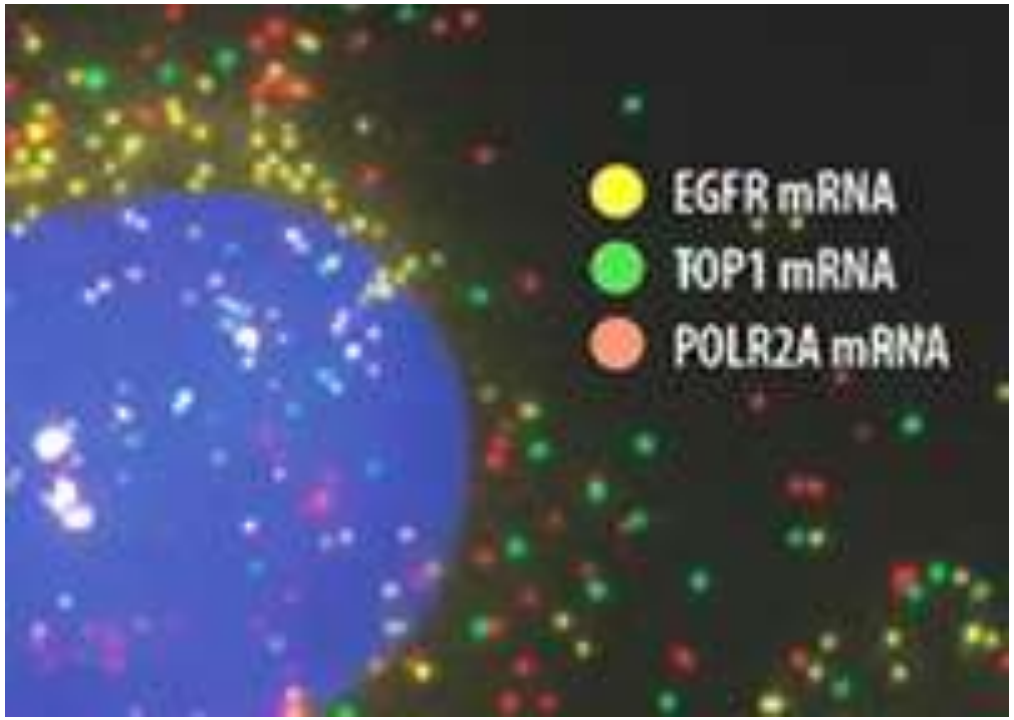
(Perea et al. Front Cell Neurosci. 2014.)

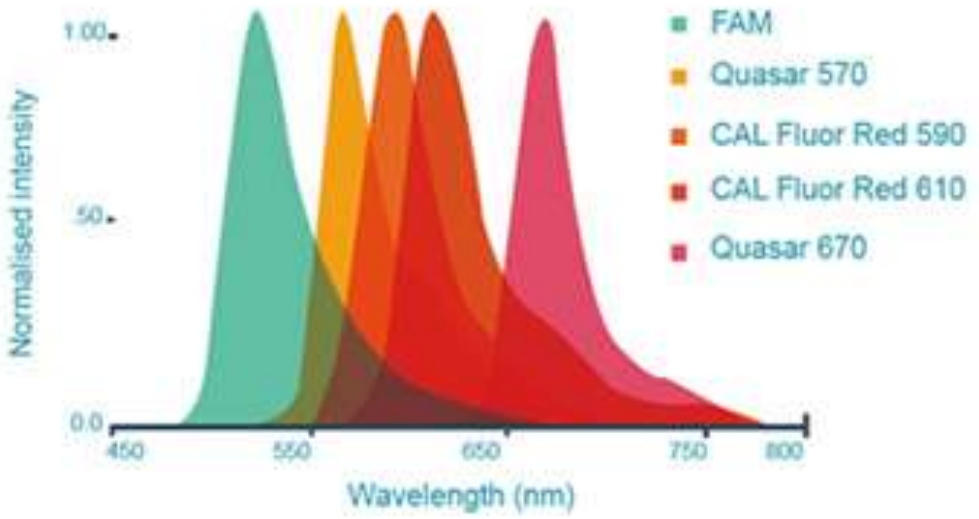
Tumor-microenvironment interaction in cancer



(Guha. The Pharmaceutical Journal. 2014.)

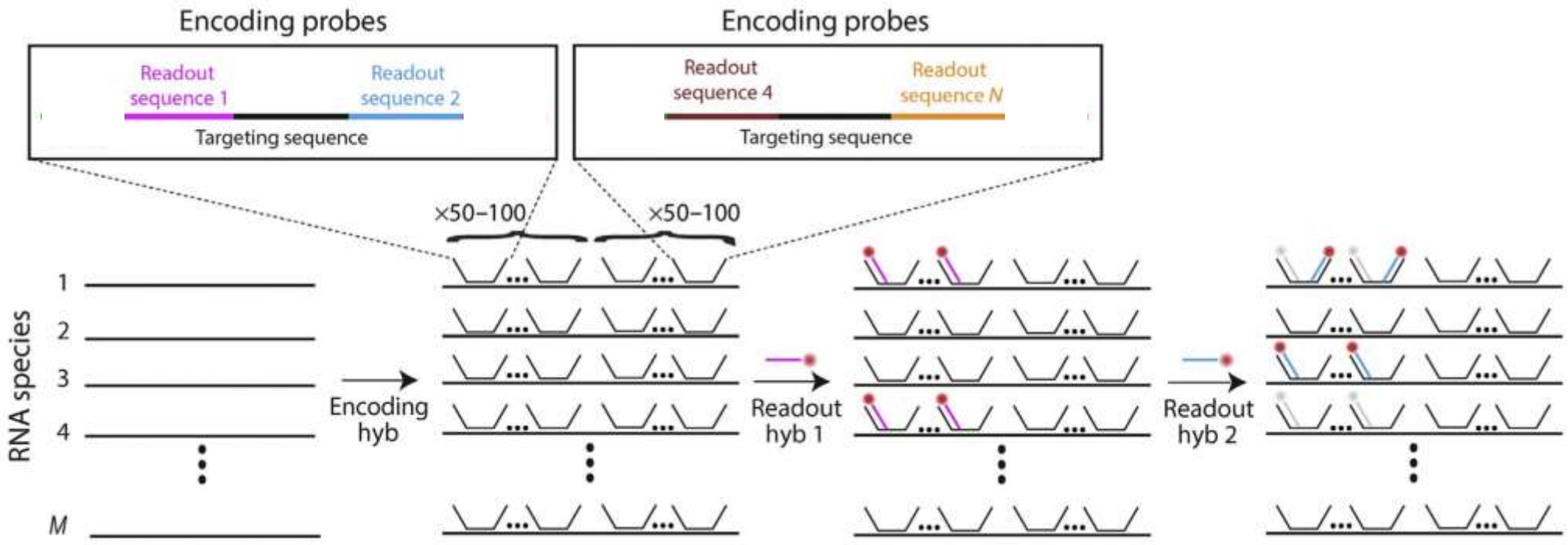
smFISH enables spatially-resolved RNA measurements at the single-cell and subcellular resolution but for very few targets due to spectral overlap





(Source: BioCat GmbH)

Multiplexed error-robust FISH (MERFISH) uses an error-robust barcoding scheme to enable simultaneous transcriptome-scale measurements



(Chen et al, Science 2015)

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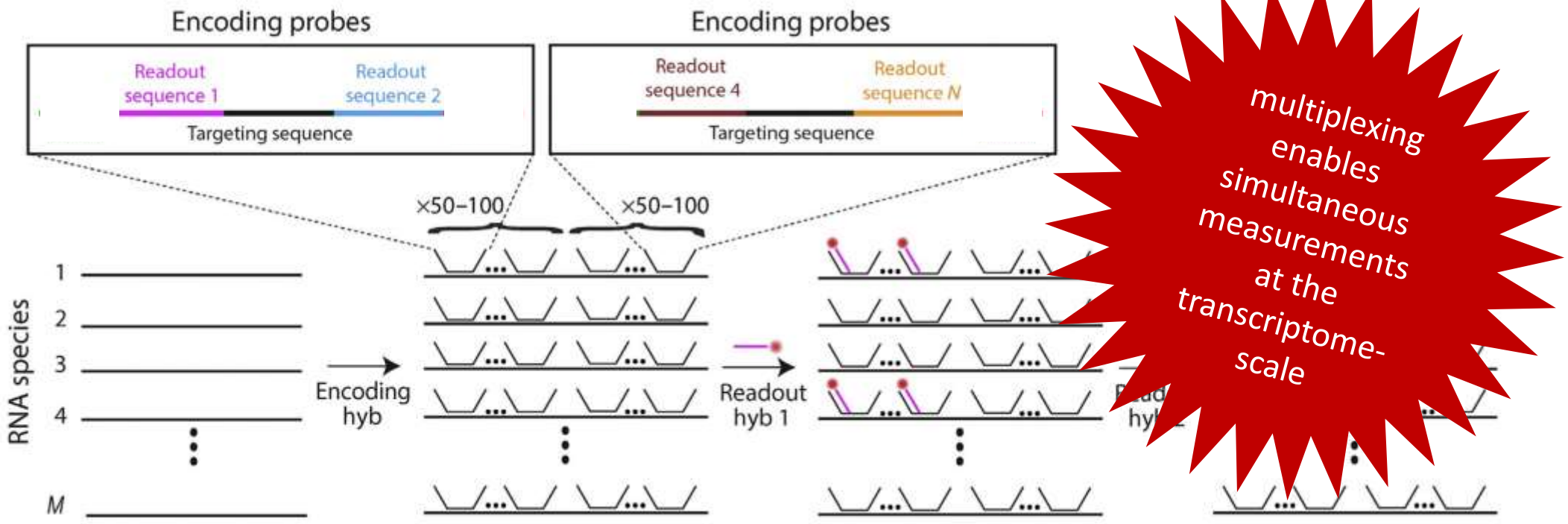
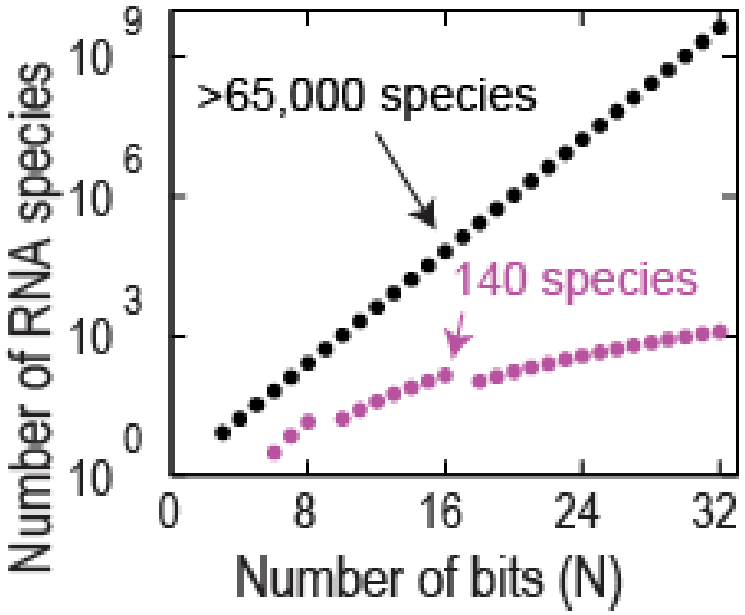
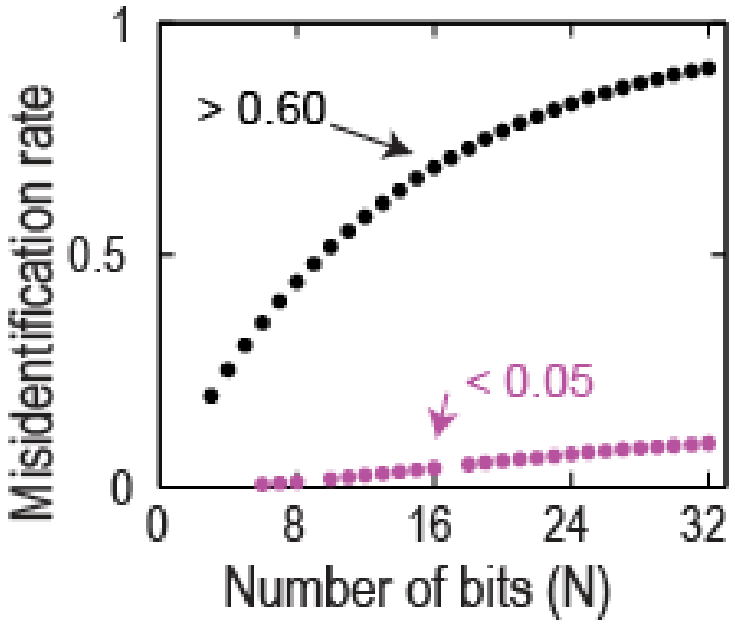


(Chen et al, Science 2015)

MERFISH uses error-robust binary codes to enable error detection and correction





- All binary combinations (2^N)
- Modified Hamming Distance 4 (MHD4) code

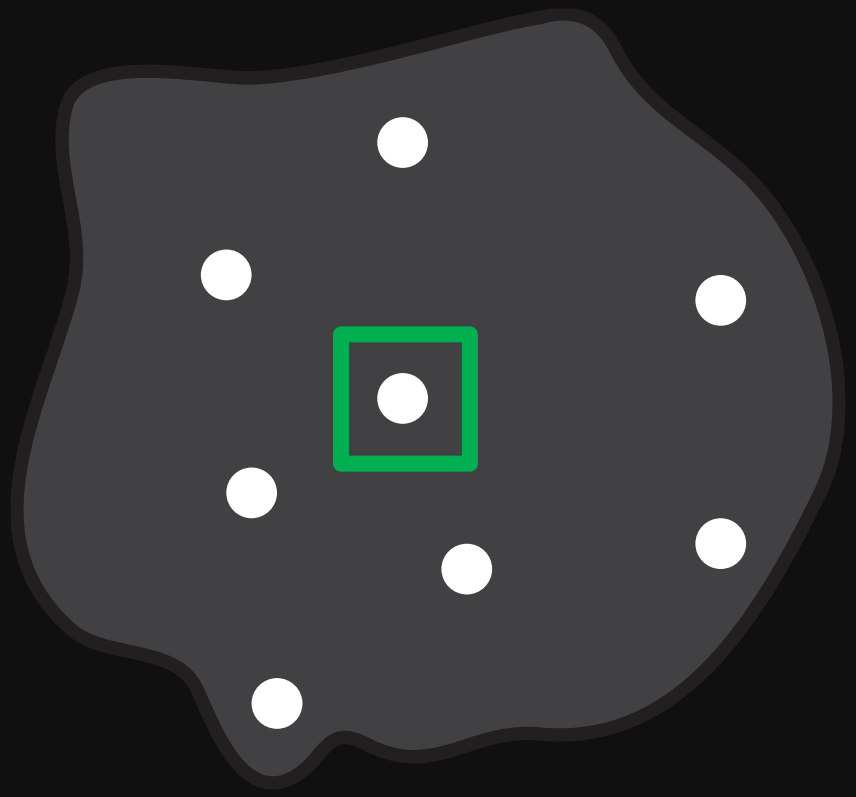
Hamming Code, Distance 4

Gene A: 1 1 0 0 0 0 0 1 0 0 0 0 0 0 1 0

Gene B: 0 1 0 0 0 0 0 1 0 0 0 1 0 0 0 1

Simulated Raw Data

Image 1



Decoded barcode:
1

3 bit barcode

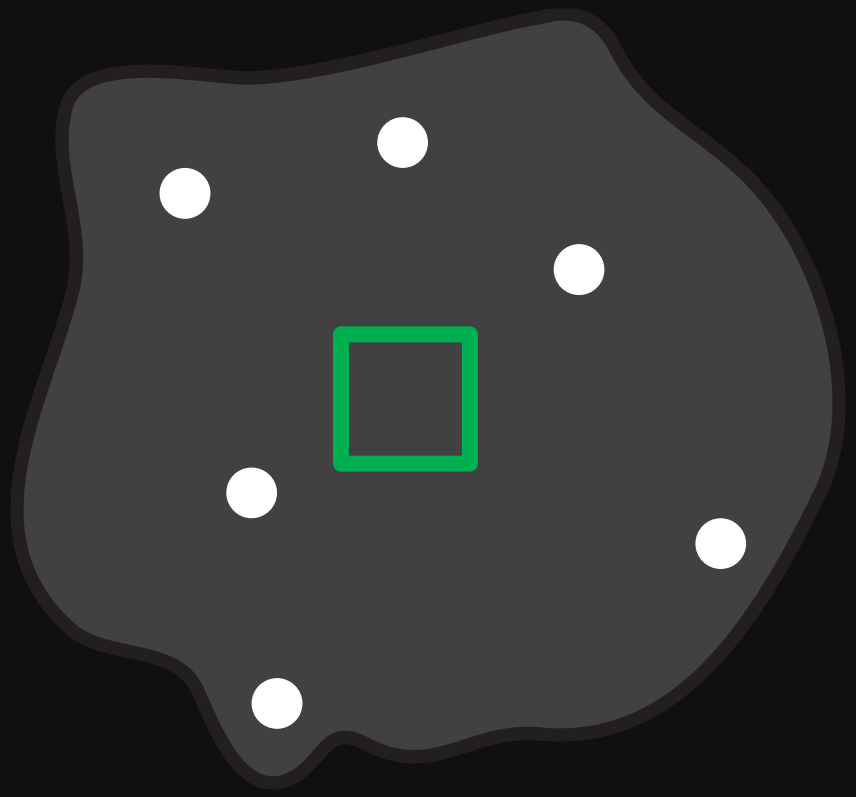
…

Gene X : 101
Gene Y : 011

…

Simulated Raw Data

Image 2



Decoded barcode:
10

3 bit barcode

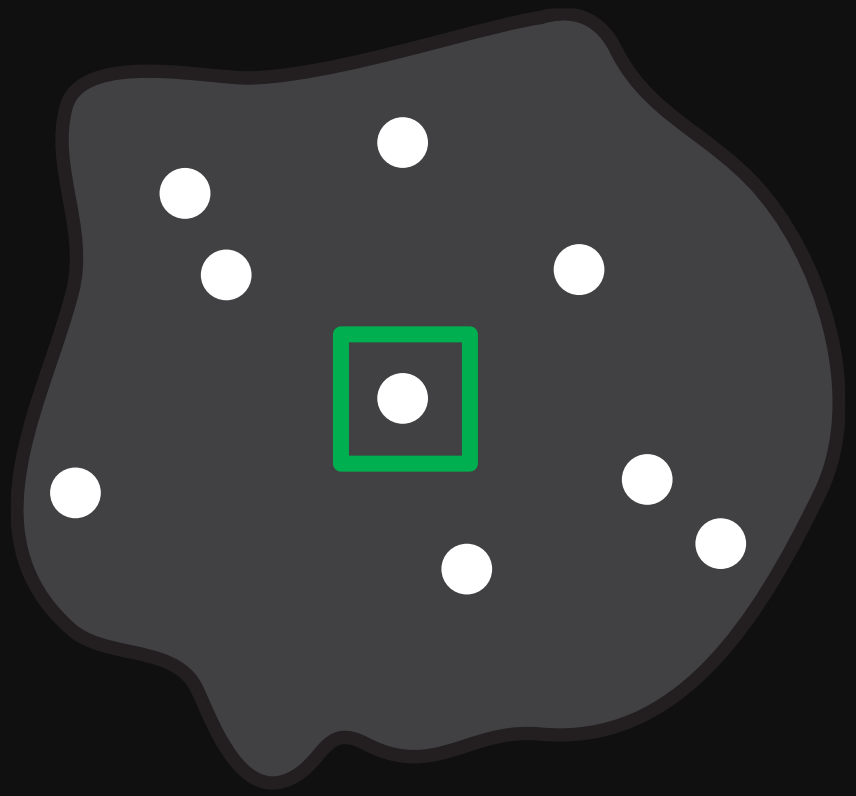
…

Gene X : 101
Gene Y : 011

…

Simulated Raw Data

Image 3



Decoded barcode:
101

3 bit barcode

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Gene X : 101
Gene Y : 011

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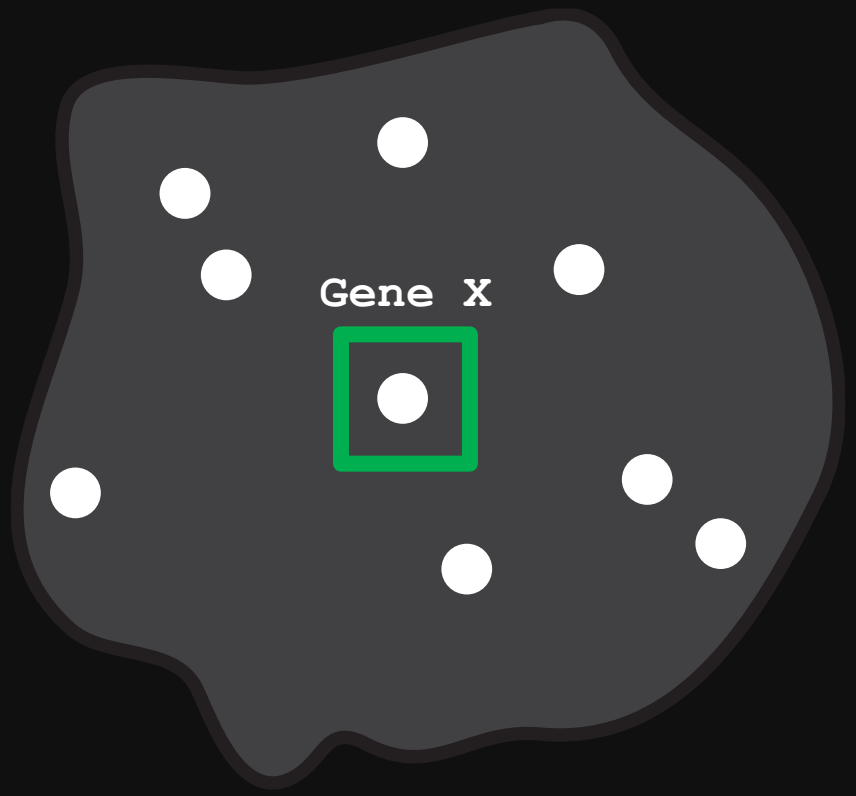
Simulated Raw Data

Image 3



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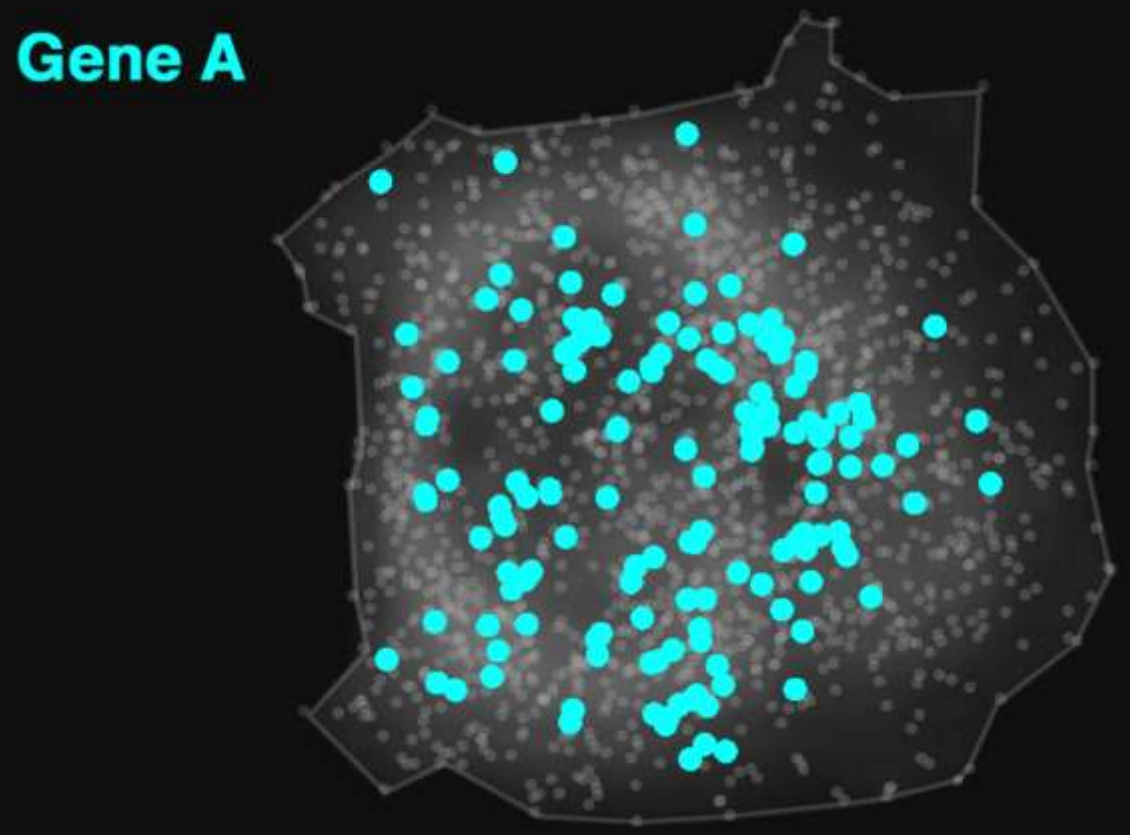
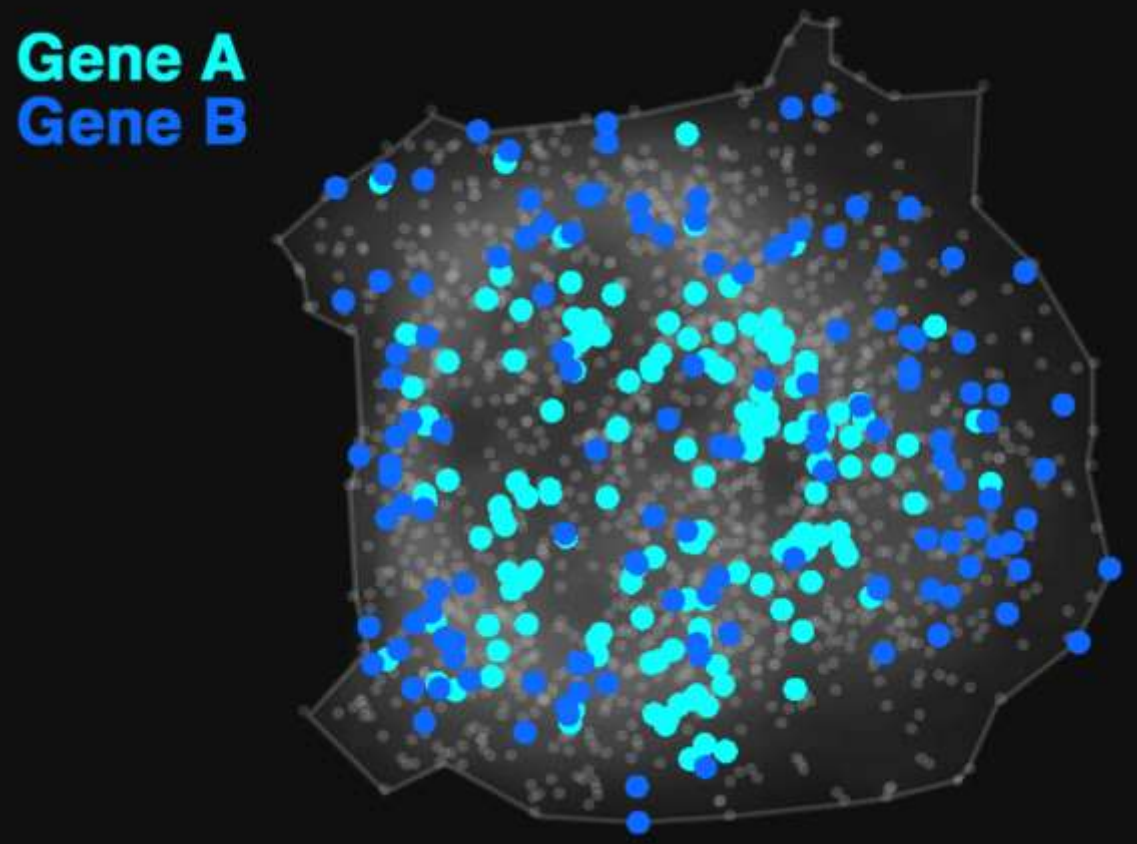


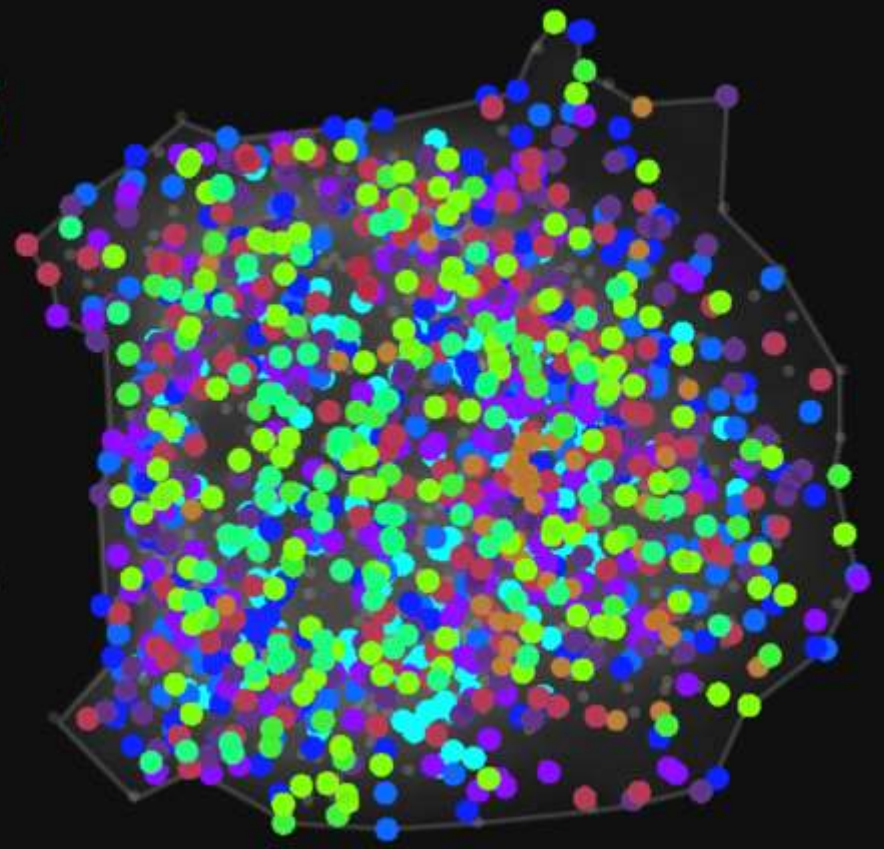
Decoded Data



Decoded Data

(Chen et al, Science 2015)
(Wang et al, Sci Rep 2018)

Gene A
Gene B
...
100s
to
~10000
of gene
species



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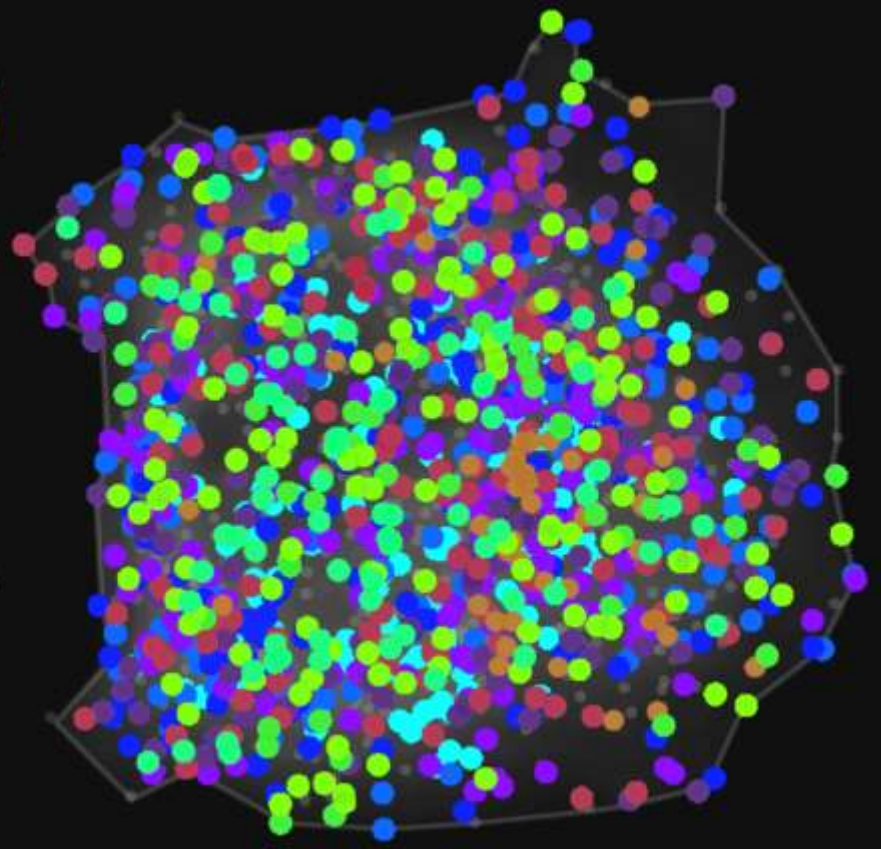
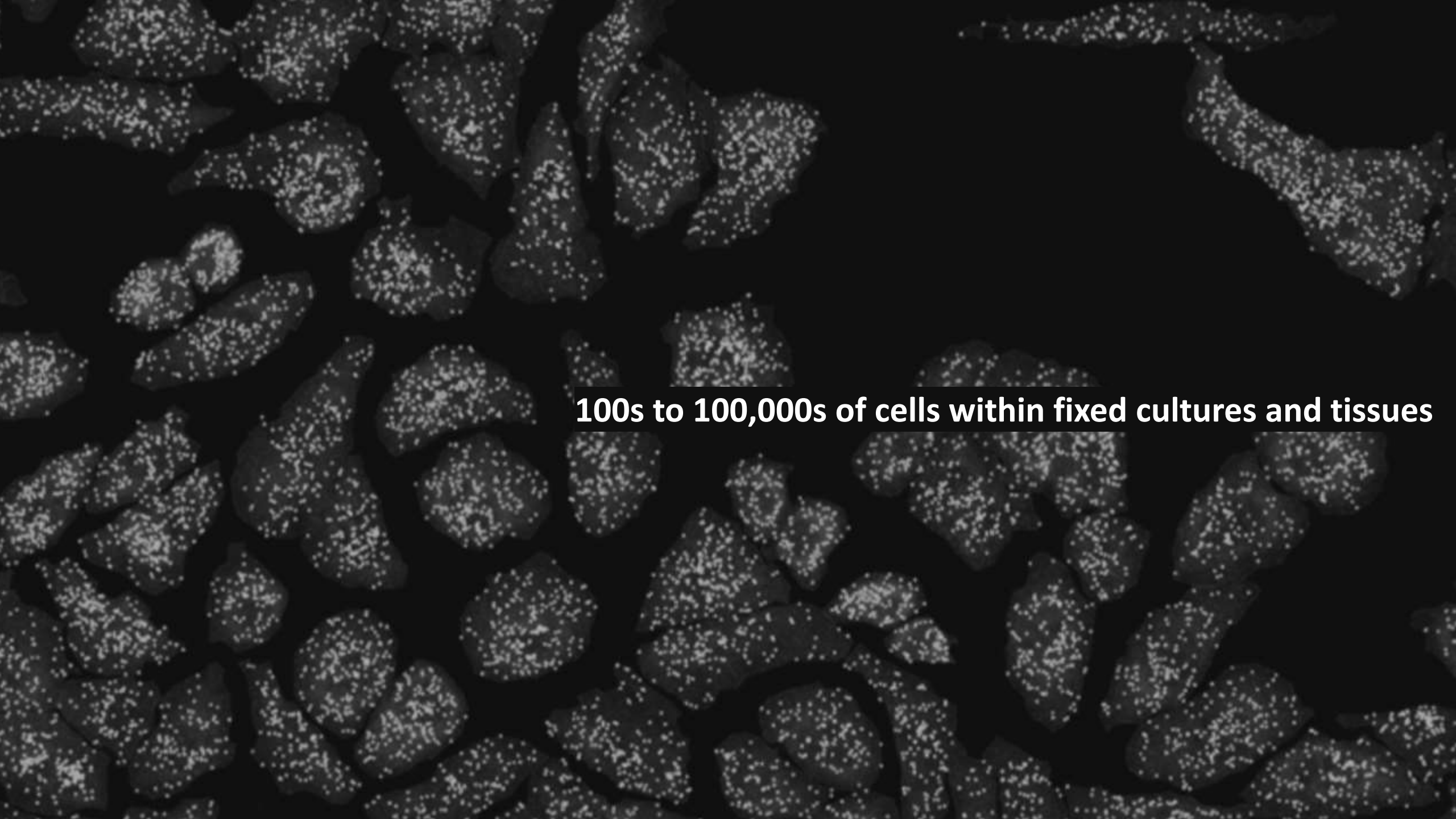
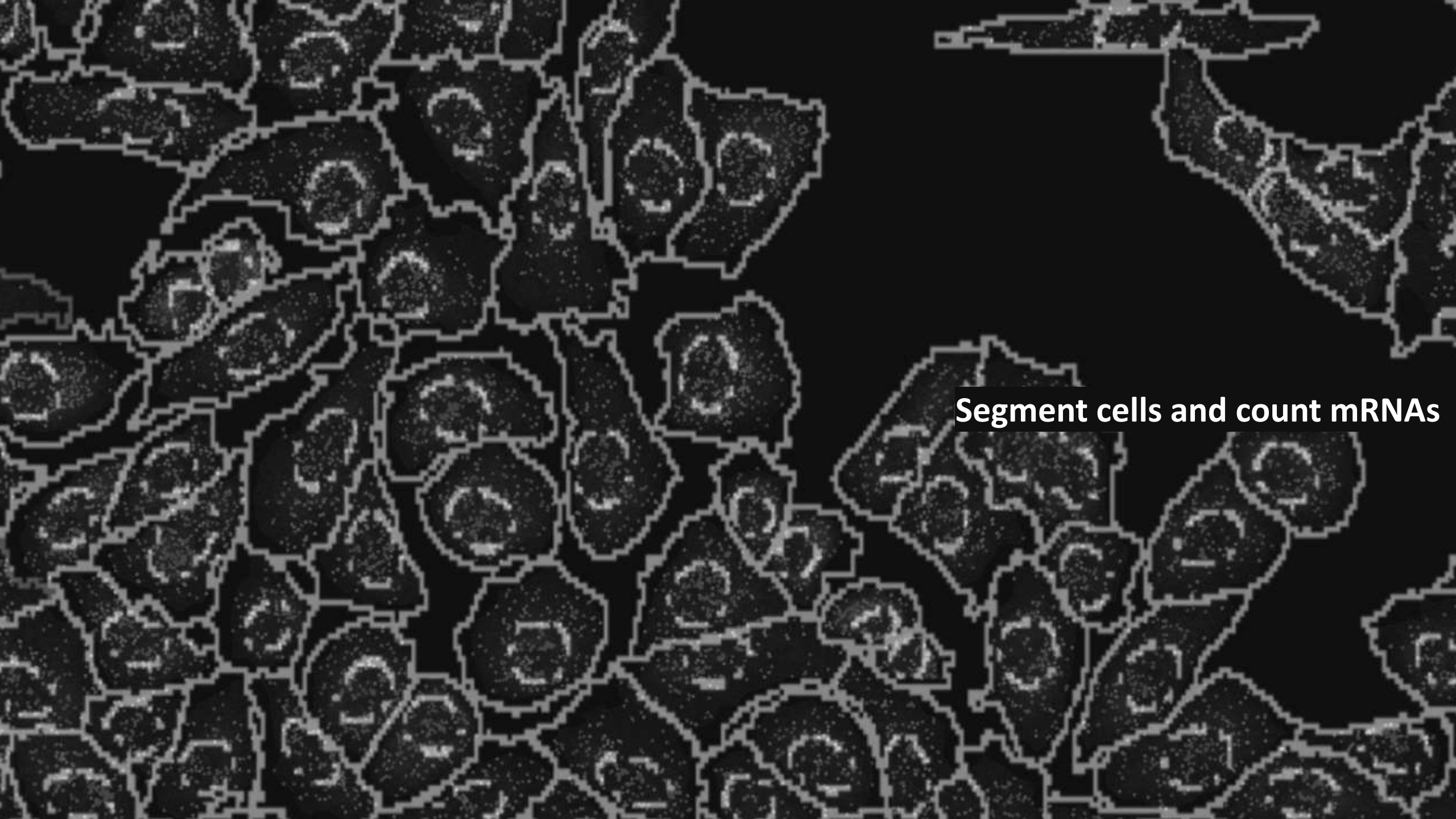
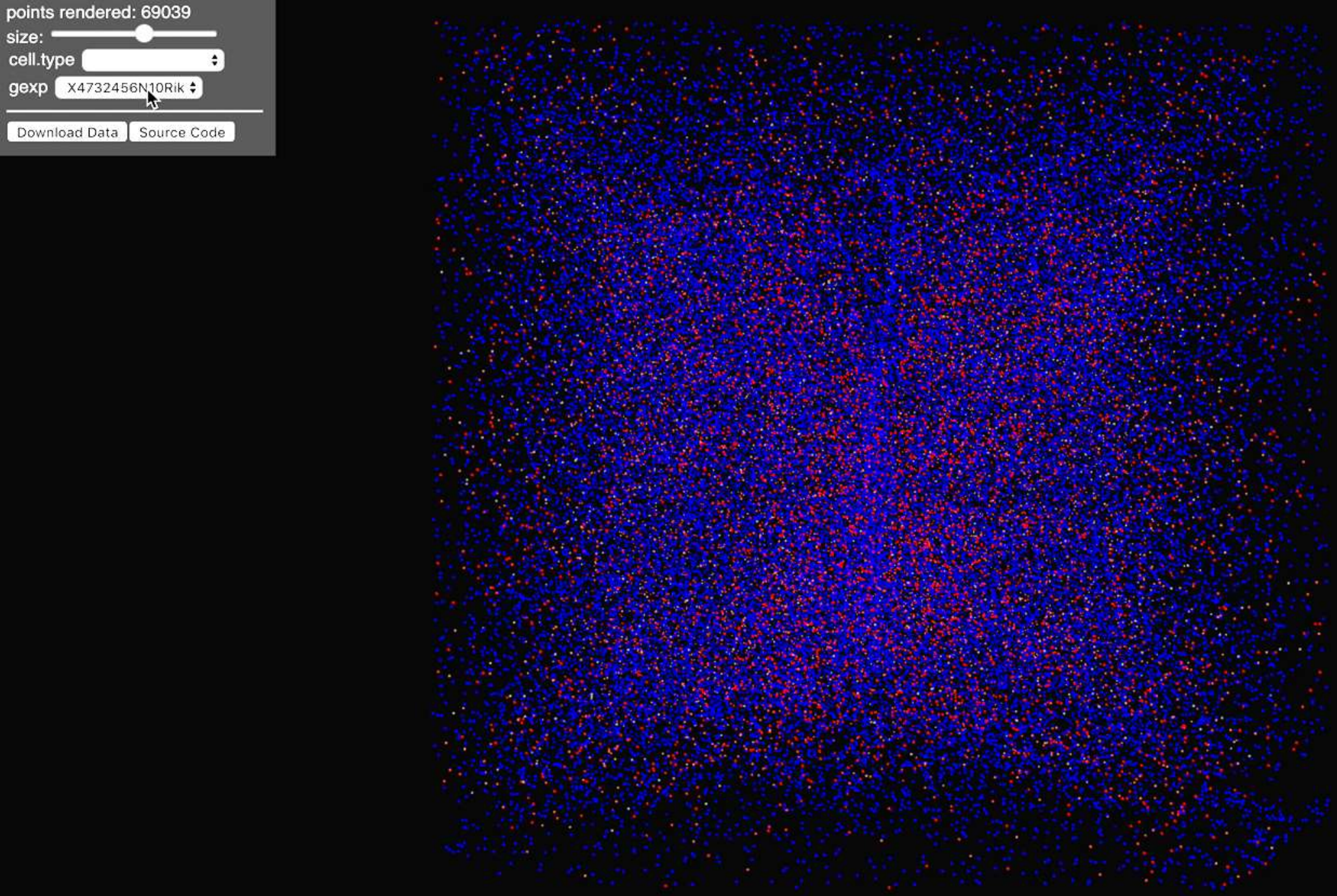


~80% detection efficiency
~4% mis-identification


100s to 100,000s of cells within fixed cultures and tissues


Segment cells and count mRNAs

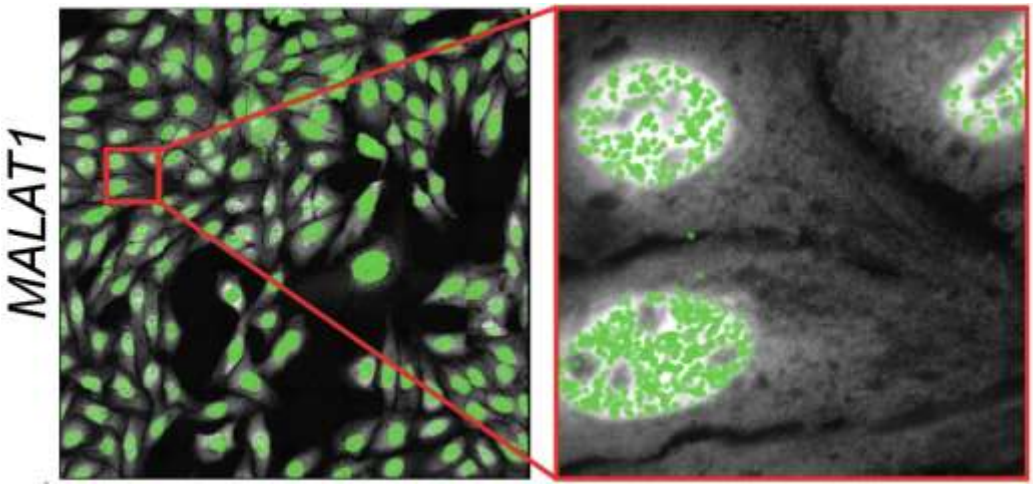


MERmaid

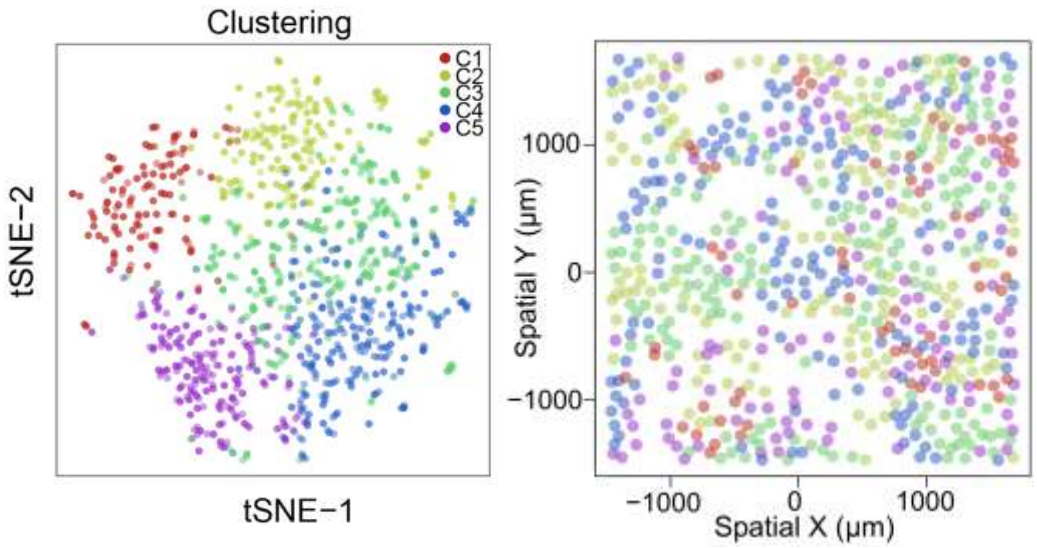
a React WebGL Tool for Exploring Spatially Resolved Single-Cell Transcriptomics Data

Such high-throughput spatially-resolved measurements combined with novel computational analyses enable addressing many biological questions

Subcellular localization of RNAs



Spatial organization of single-cell clusters



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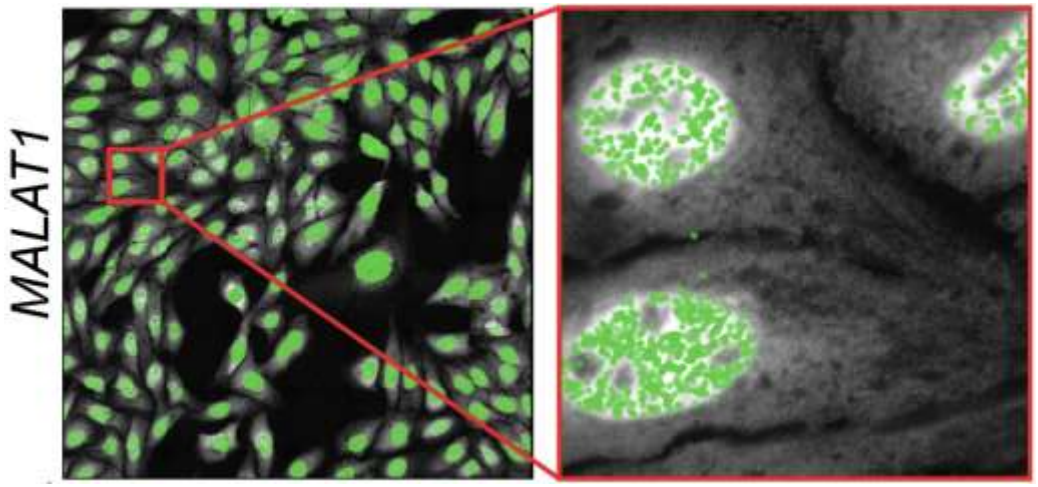

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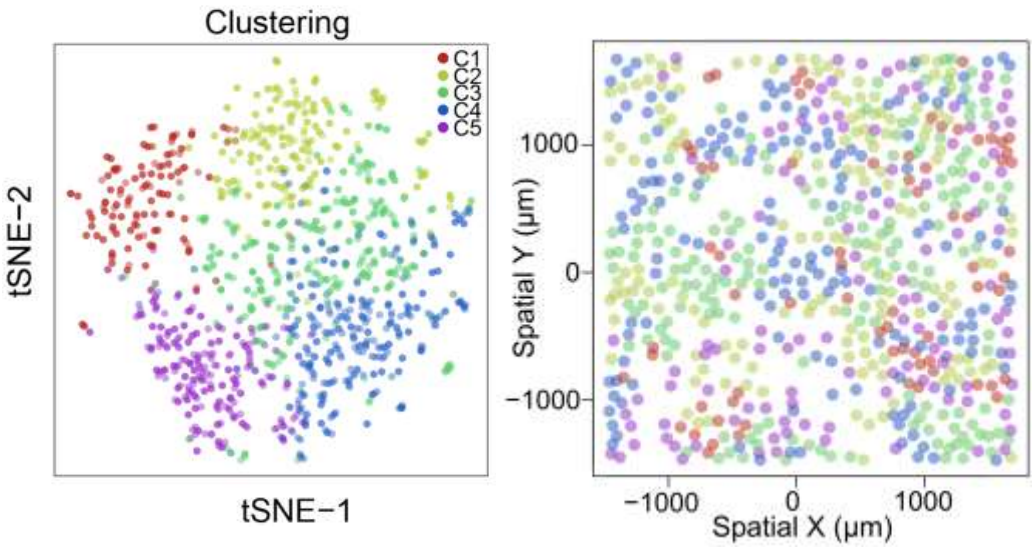
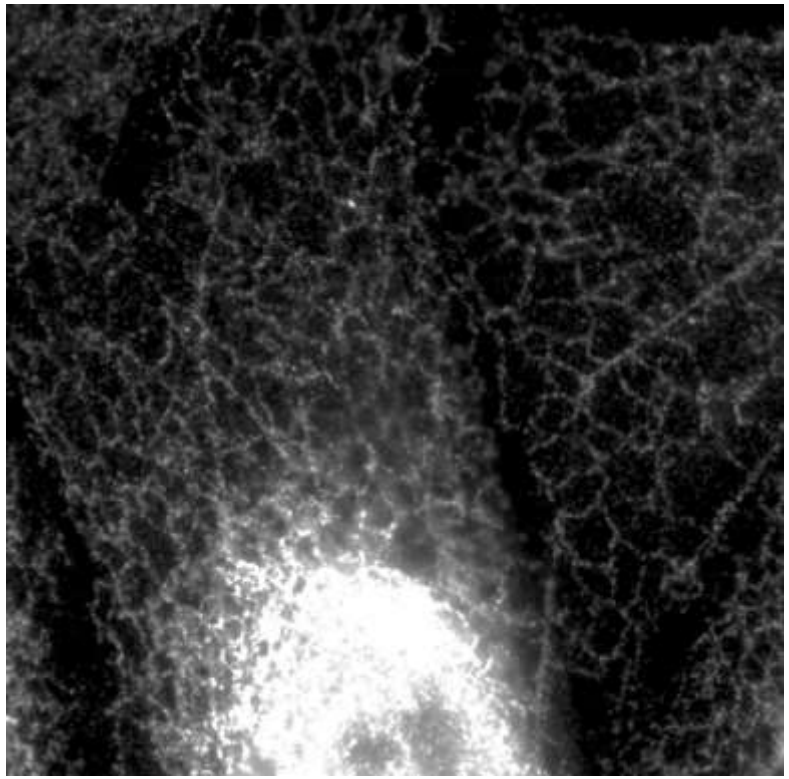
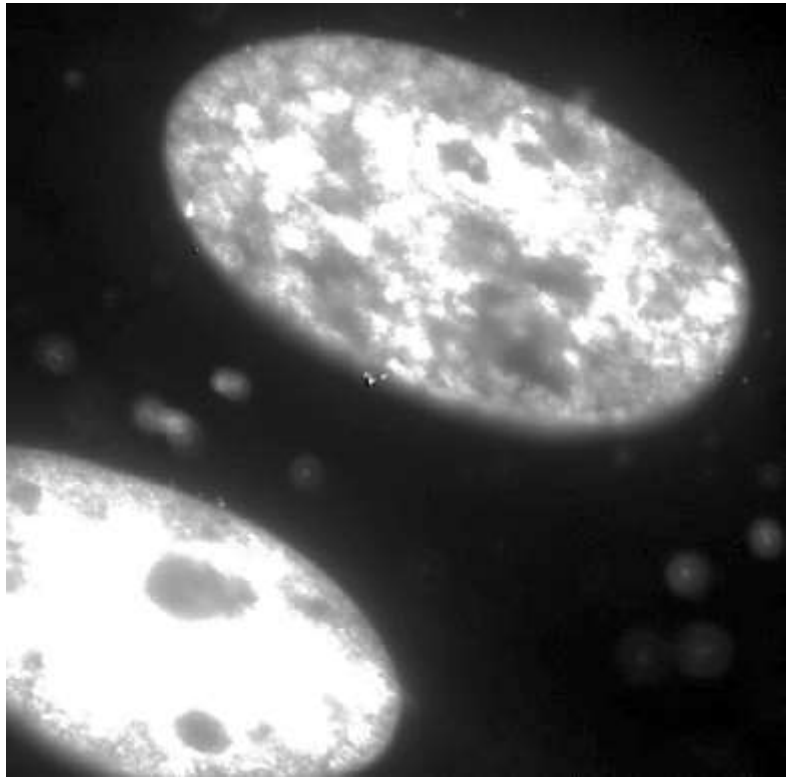


MERFISH + immunostaining + segmentation identifies genes statistically enriched at different subcellular compartments

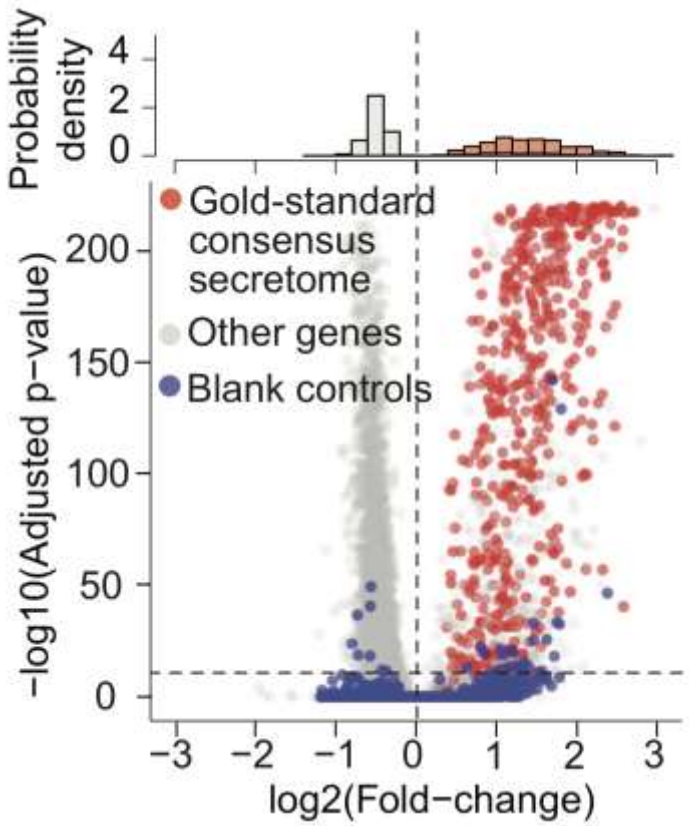
KDEL staining for ER

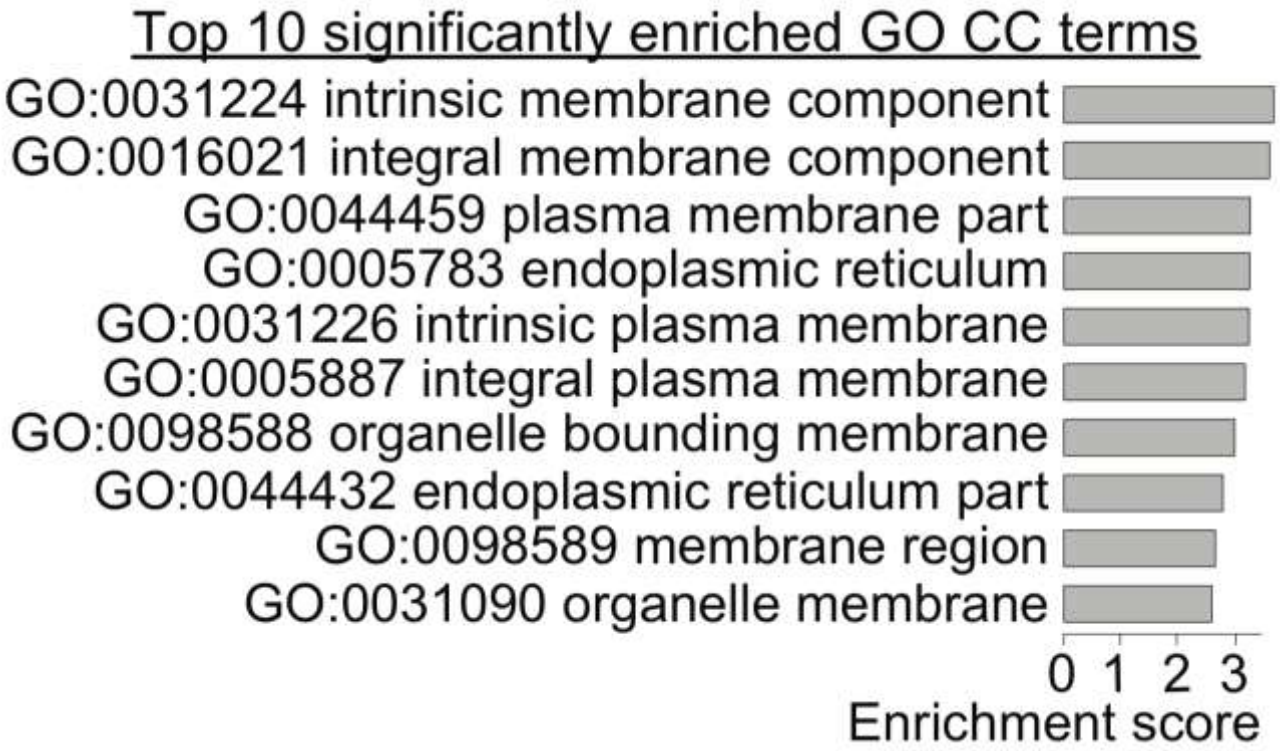


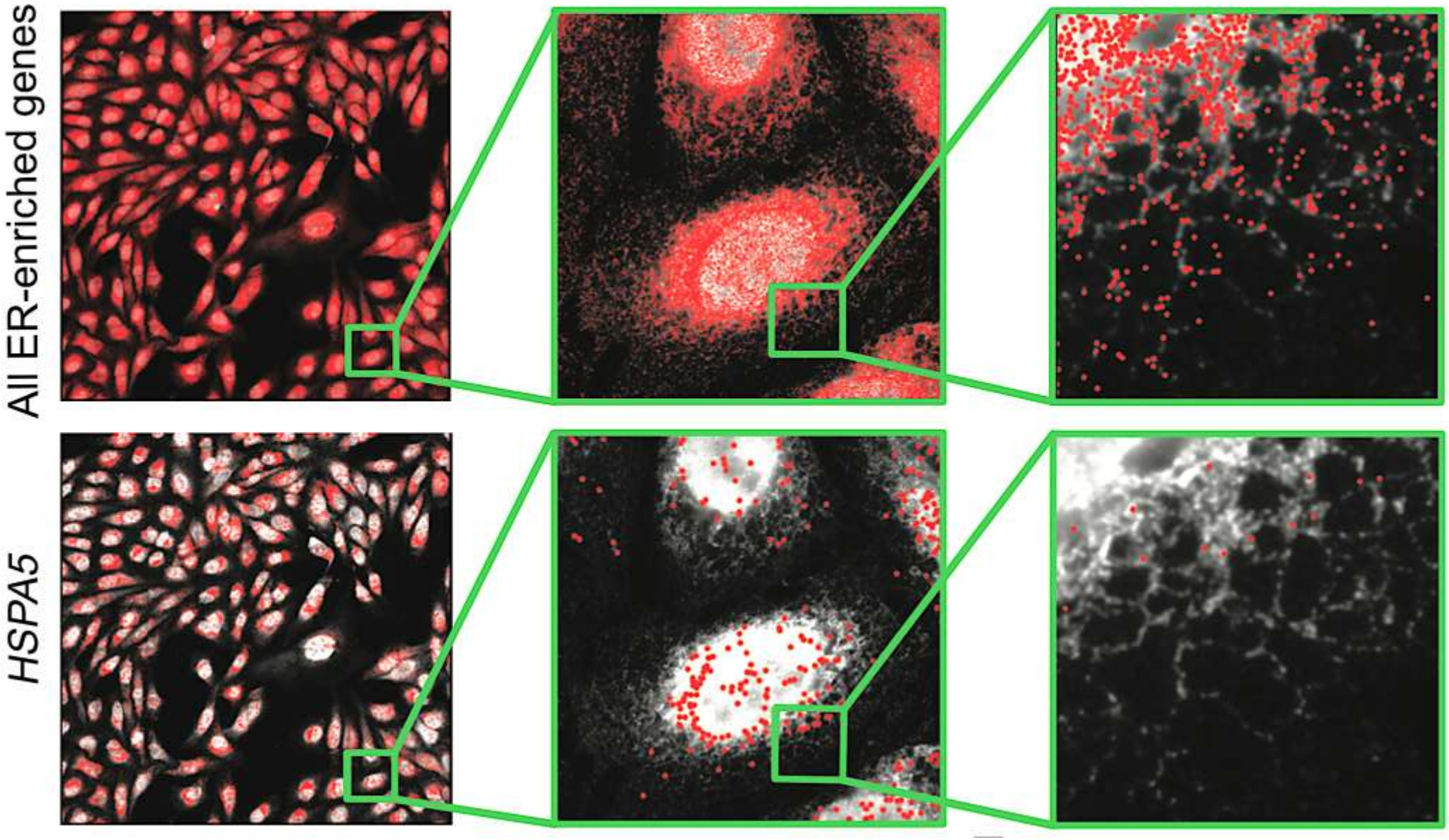
DAPI staining for nucleus



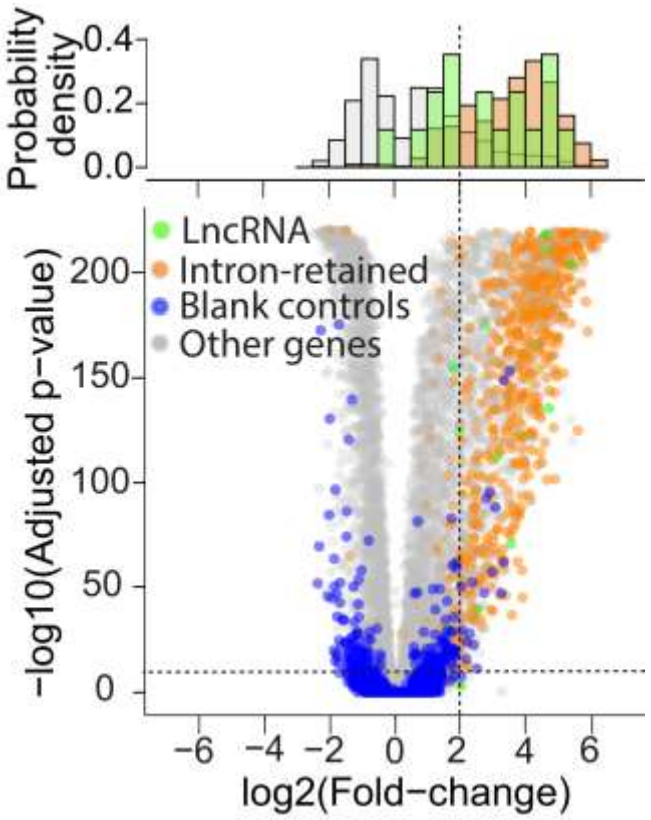
MERFISH + KDEL staining + segmentation confirms co-localization of secretome genes to the ER

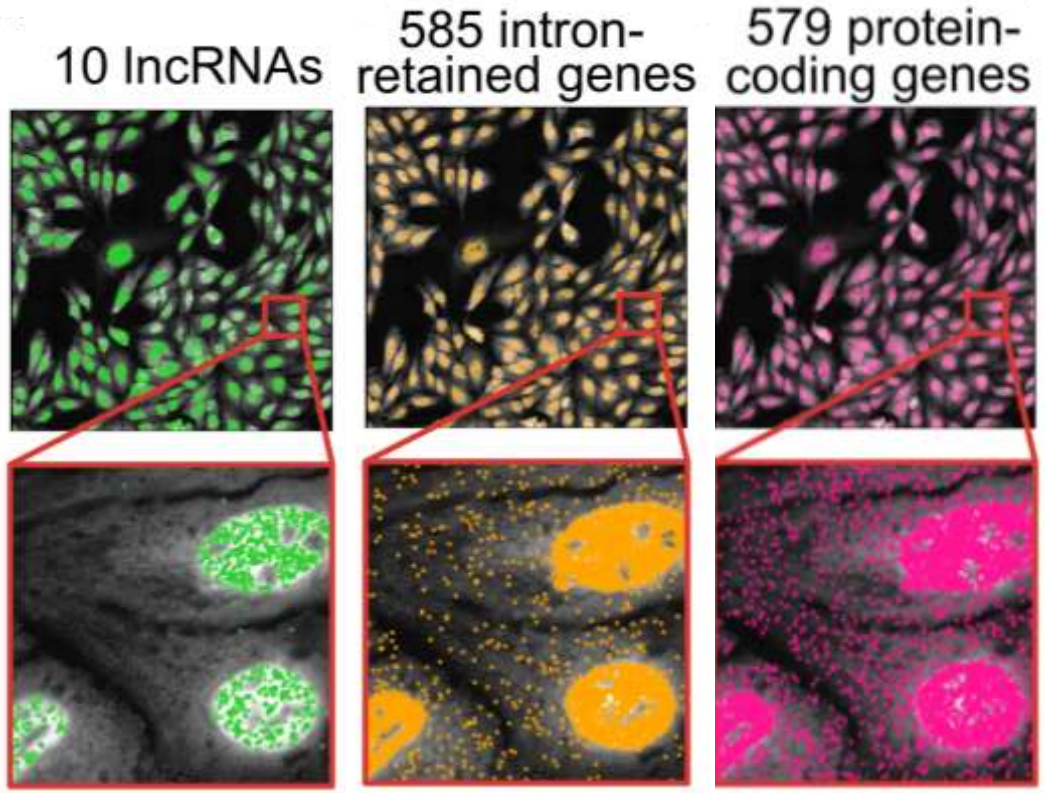


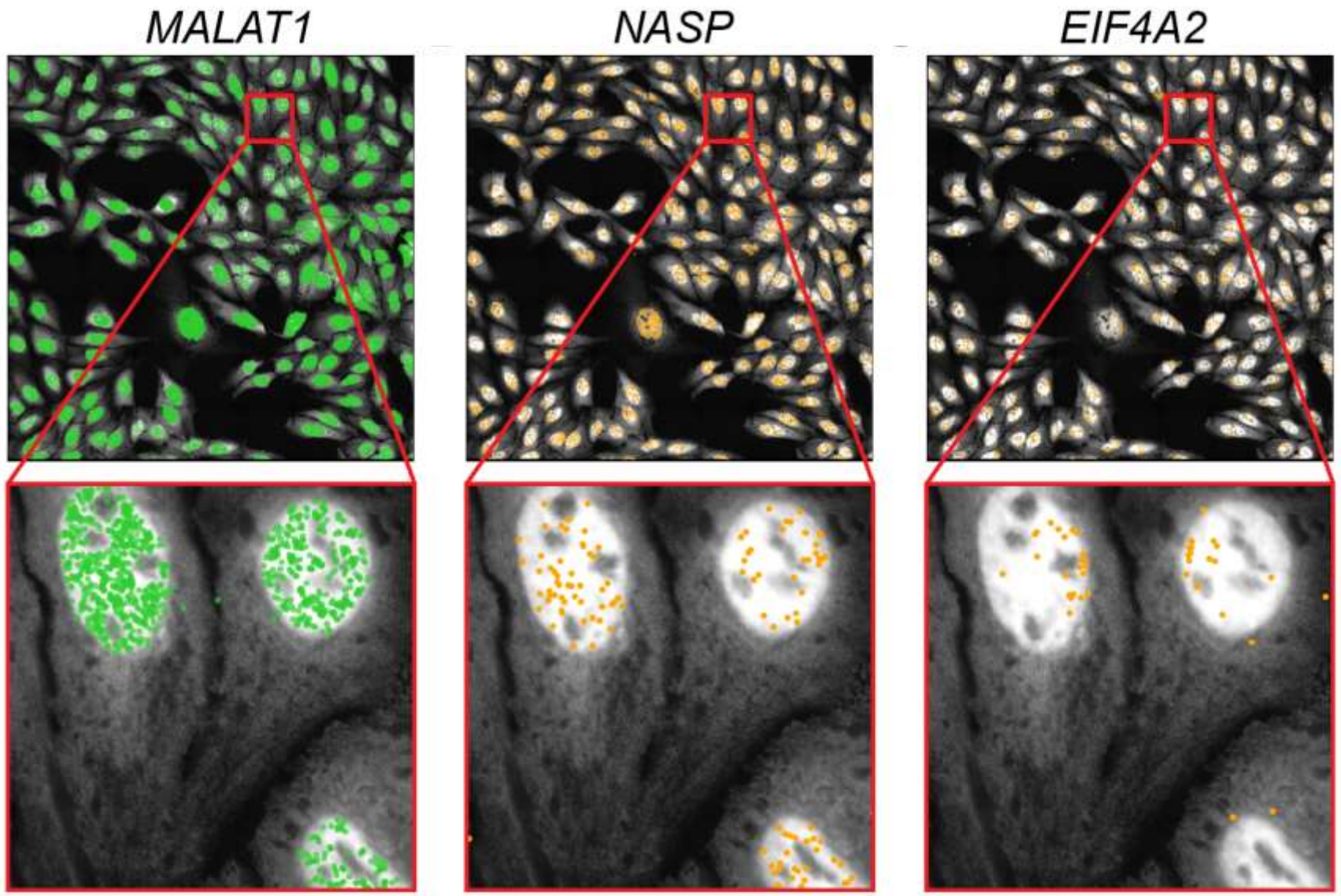



All ER-enriched genes
HSPA5

MERFISH + DAPI staining + segmentation identifies genes enriched in the nucleus

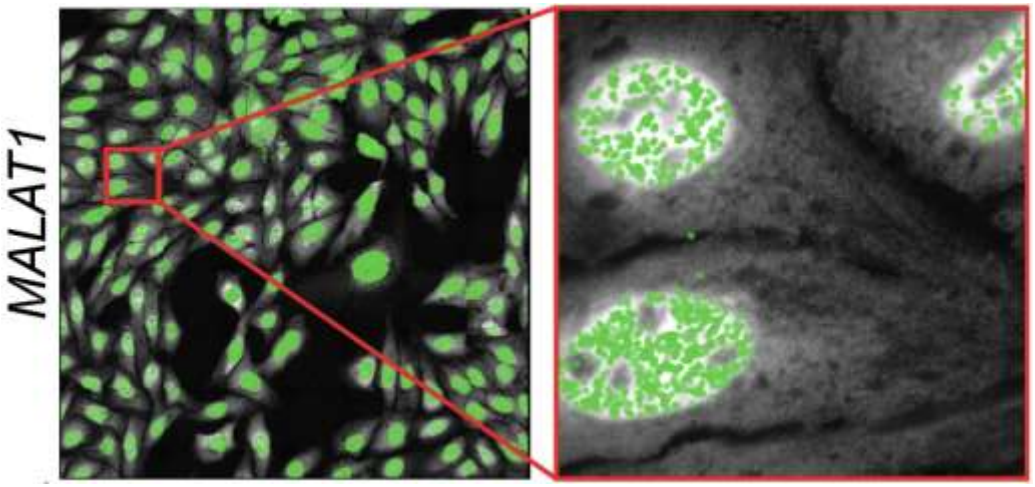




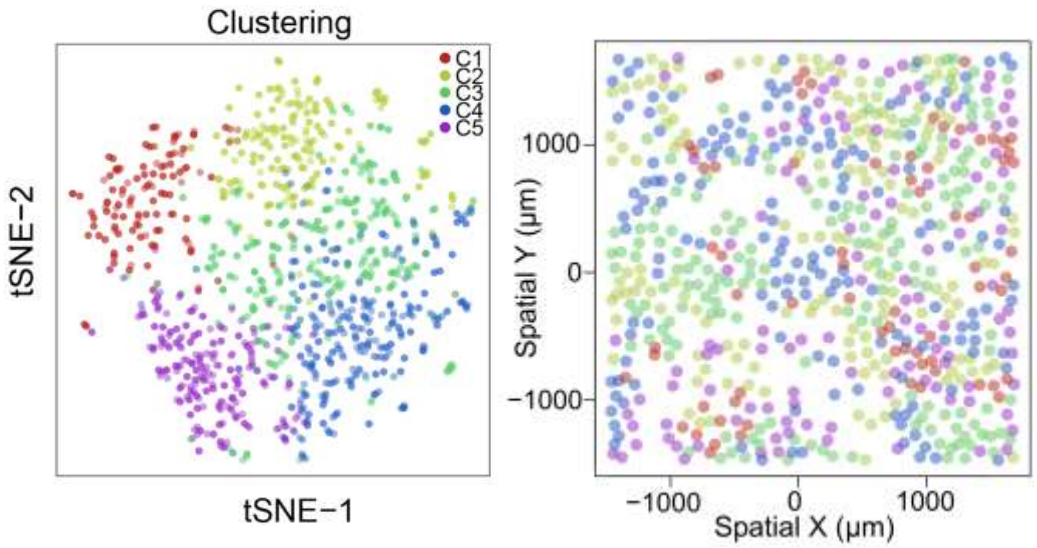

MALAT1
NASP
EIF4A2

Such high-throughput spatially-resolved measurements combined with novel computational analyses enable addressing many biological questions

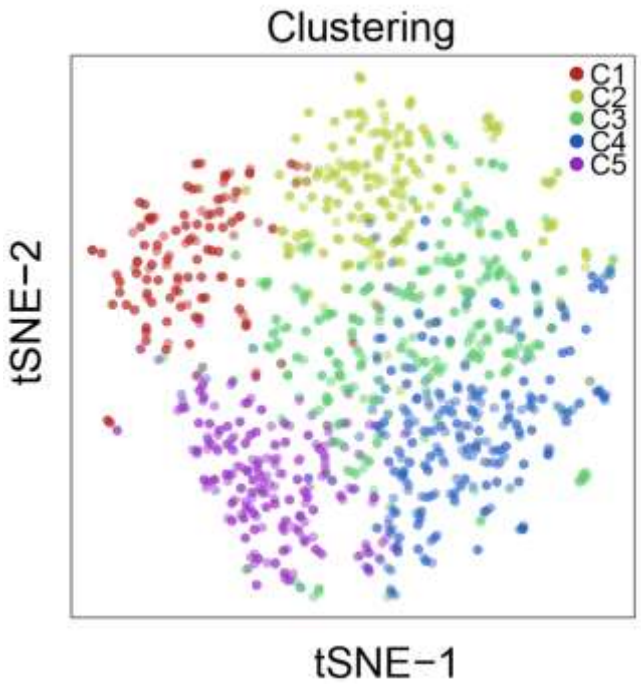
Subcellular localization of RNAs



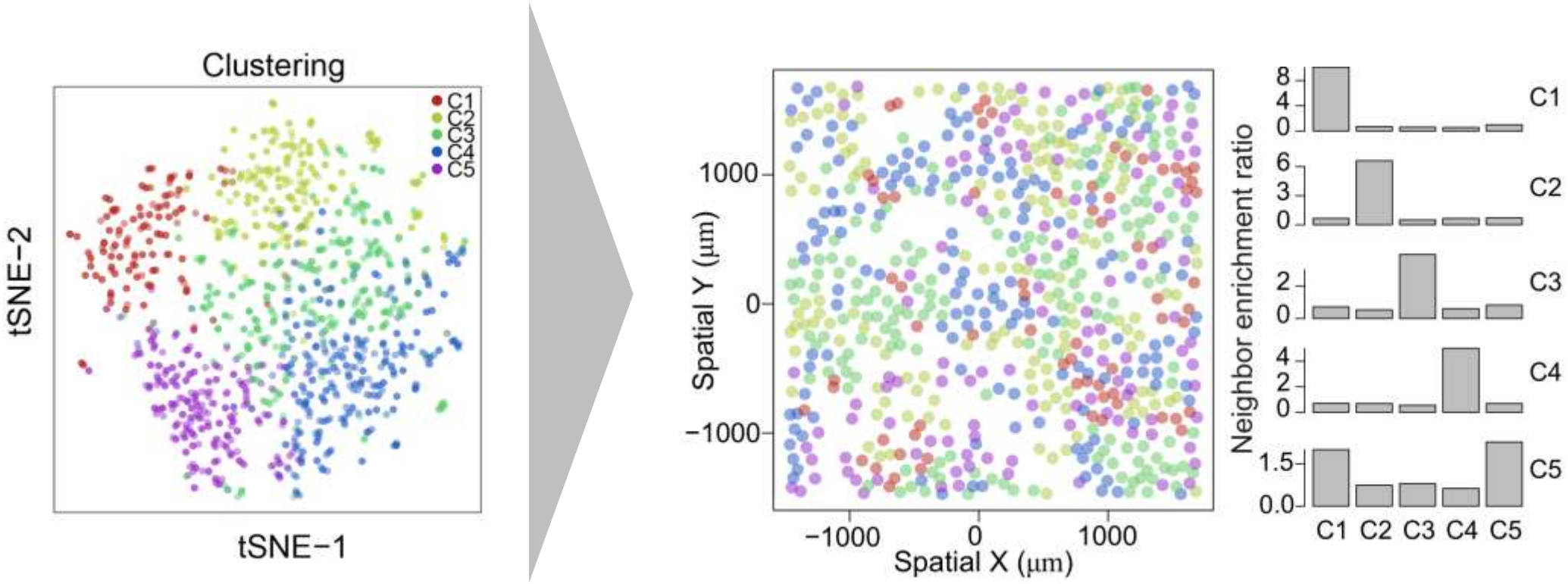
Spatial organization of single-cell clusters



Just like in scRNA-seq analysis, single-cell clustering analysis identifies transcriptionally-distinct clusters



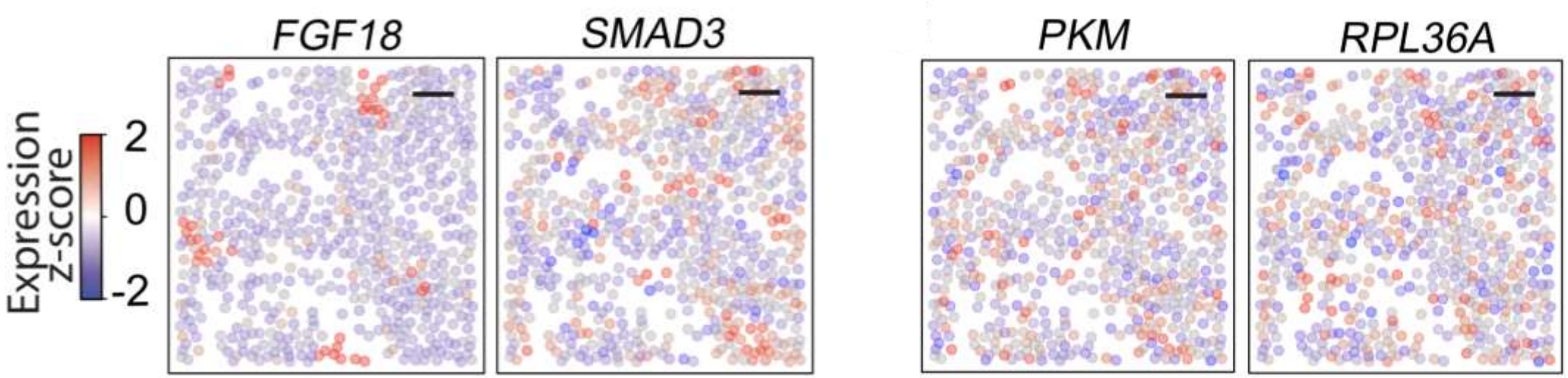
Spatial information reveals cells within each transcriptionally-distinct cluster tend to be spatially proximal to cells of the same cluster



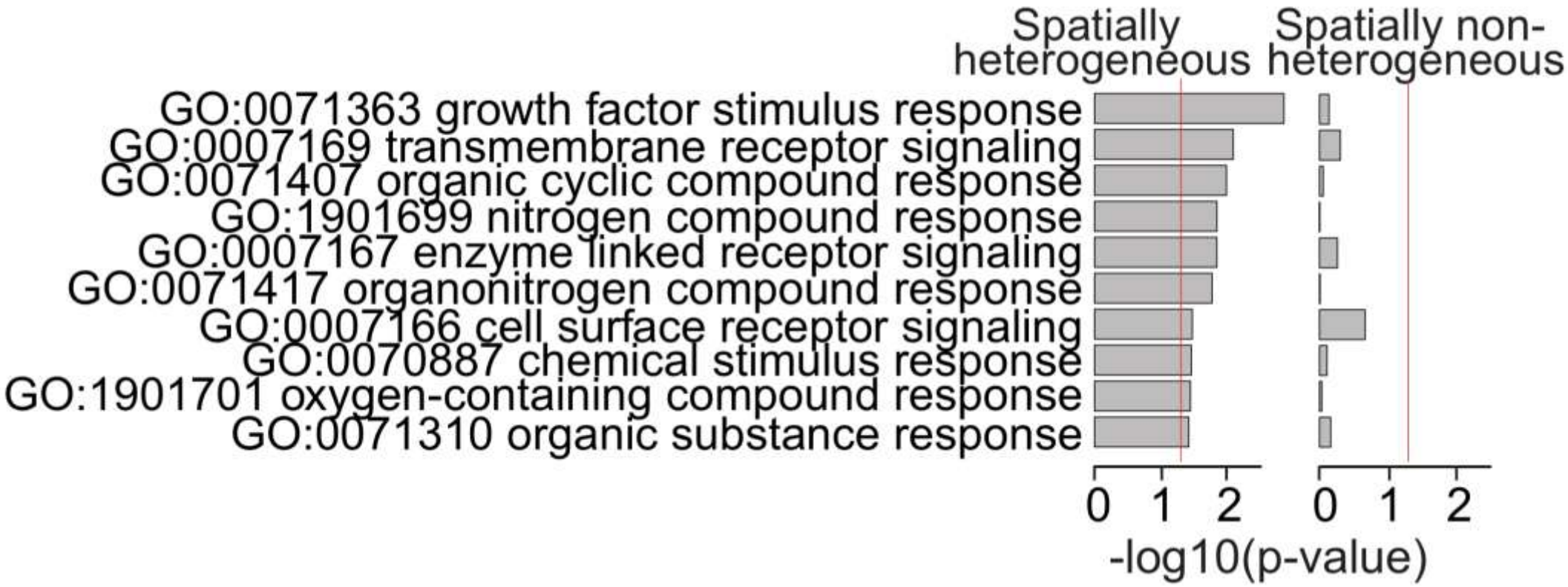
Spatially heterogeneous genes can be identified using Moran's I, a measurement of spatial clustering

Spatially heterogeneous

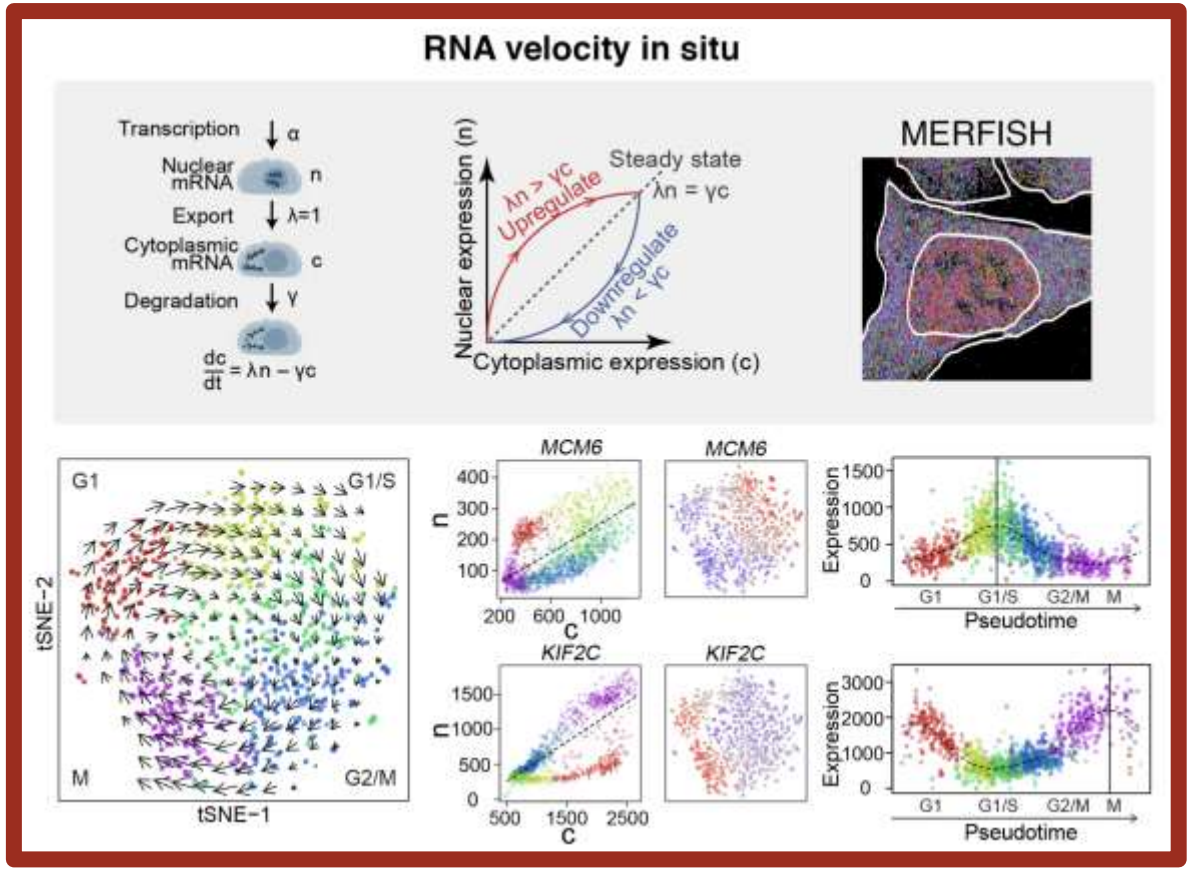
Spatially non-heterogeneous



Spatially heterogeneous genes tend to be enriched in cellular communication processes



MERFISH + computational modeling enables derivation of RNA velocity in situ



Background: Kharchenko and Linnarson lab previously demonstrated inference of RNA velocity using scRNA-seq data using the ratio of unspliced vs. spliced mRNA

nature
International journal of science

Letter | Published: 08 August 2018

RNA velocity of single cells

Gioele La Manno, Ruslan Soldatov, Amit Zeisel, Emelie Braun, Hannah Hochgerner, Viktor Petukhov, Katja Lidschreiber, Maria E. Kastriti, Peter Lönnerberg, Alessandro Furlan, Jean Fan, Lars E. Borm, Zehua Liu, David van Bruggen, Jimin Guo, Xiaoling He, Roger Barker, Erik Sundström, Gonçalo Castelo-Branco, Patrick Cramer, Igor Adameyko, Sten Linnarsson & Peter V. Kharchenko

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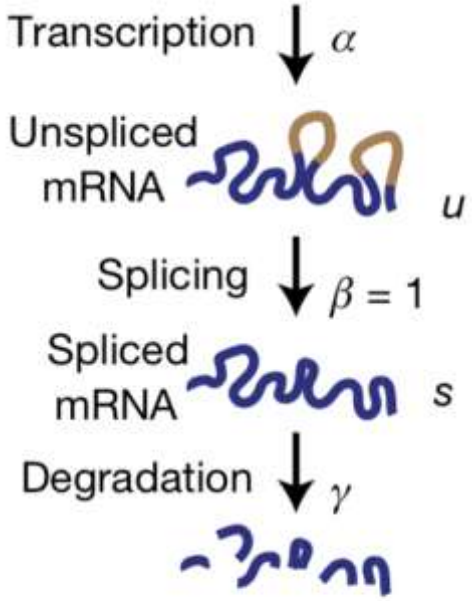

$$\frac{ds}{dt} = u - \gamma s$$

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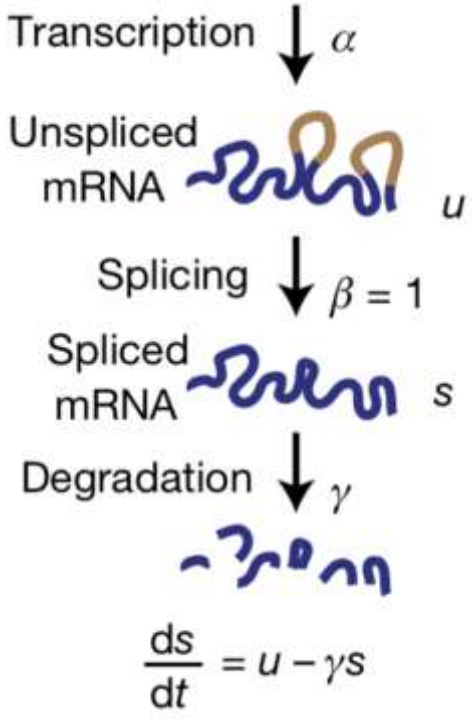

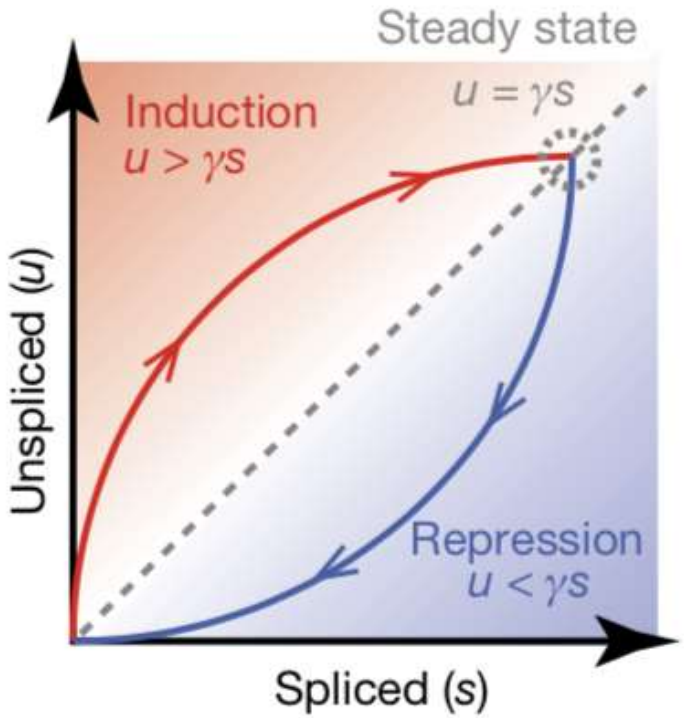

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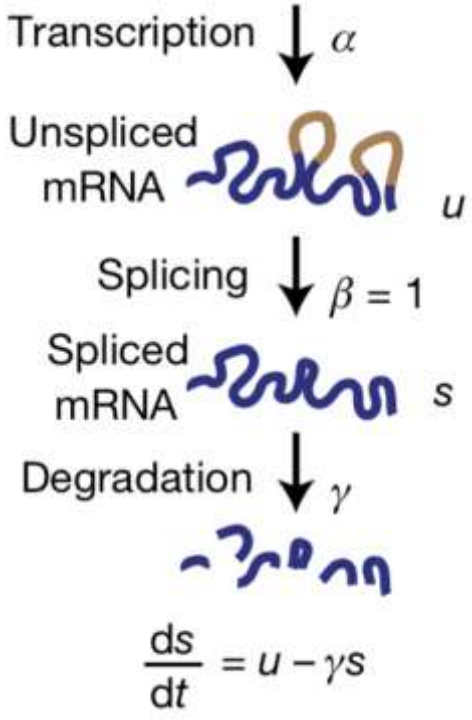

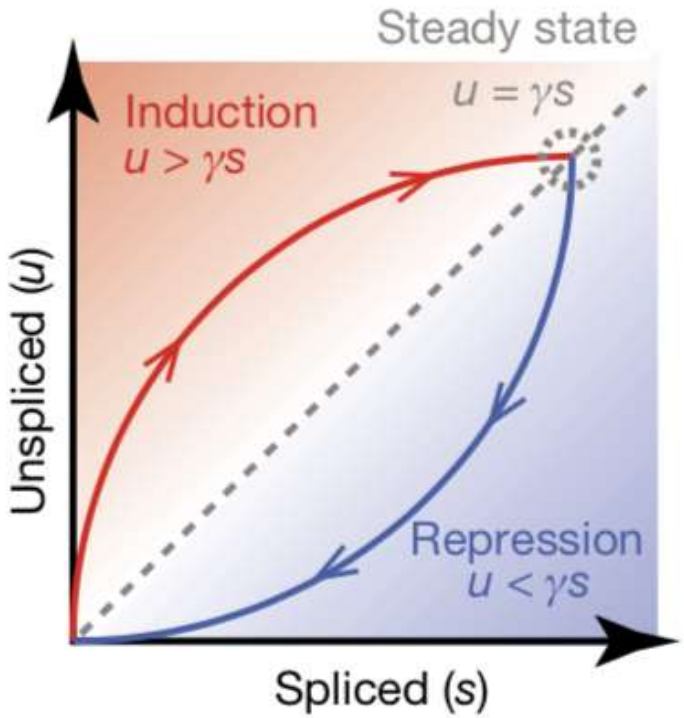

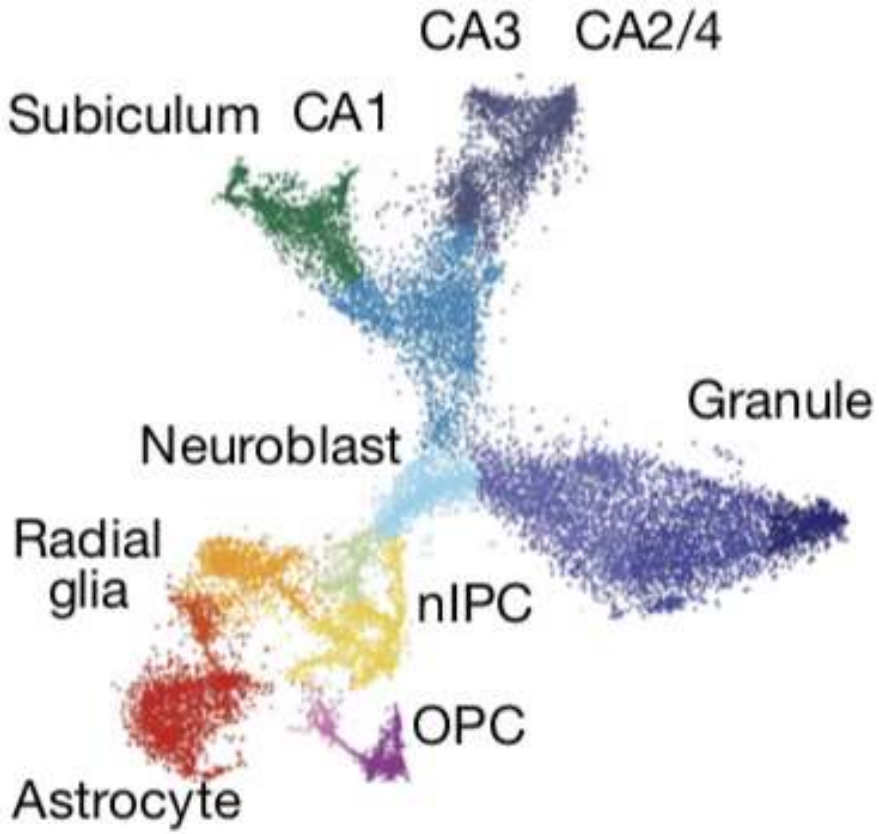

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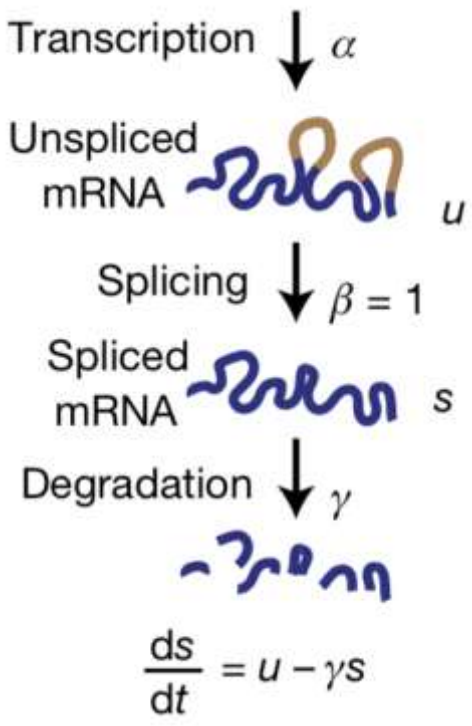
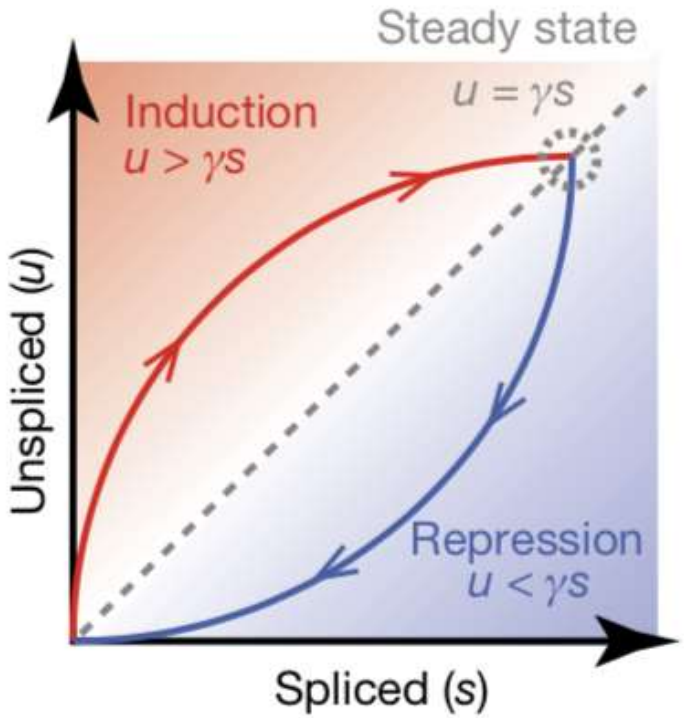
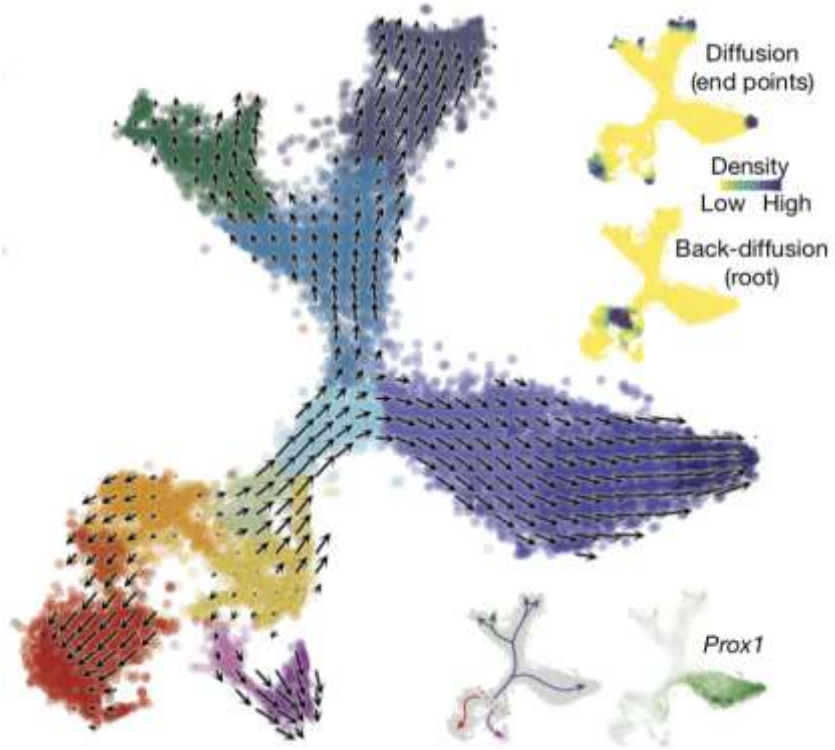
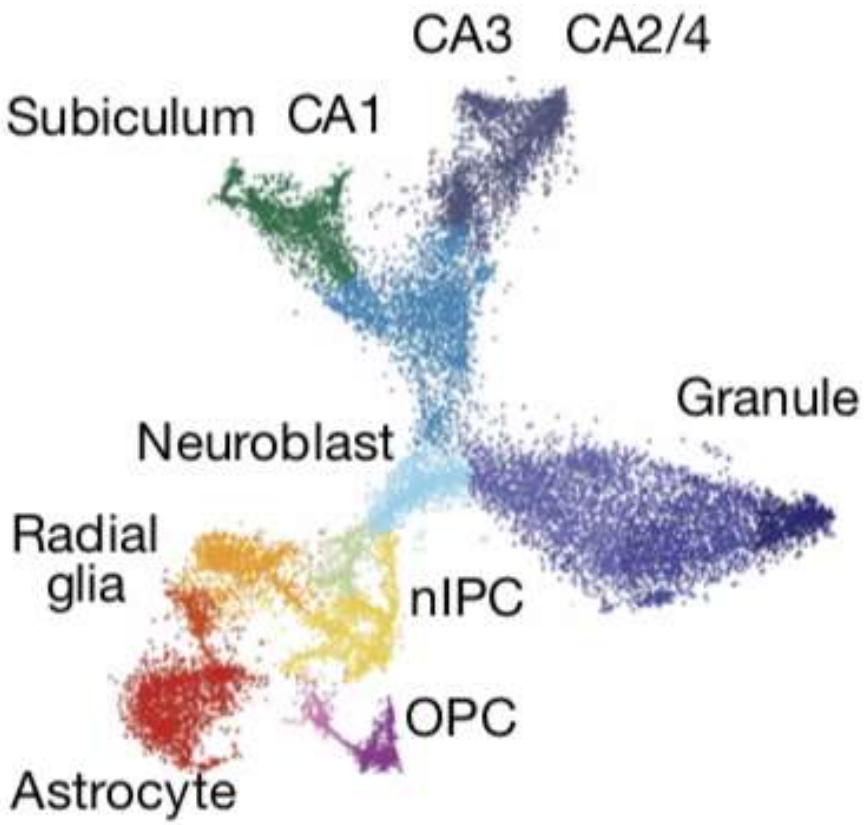
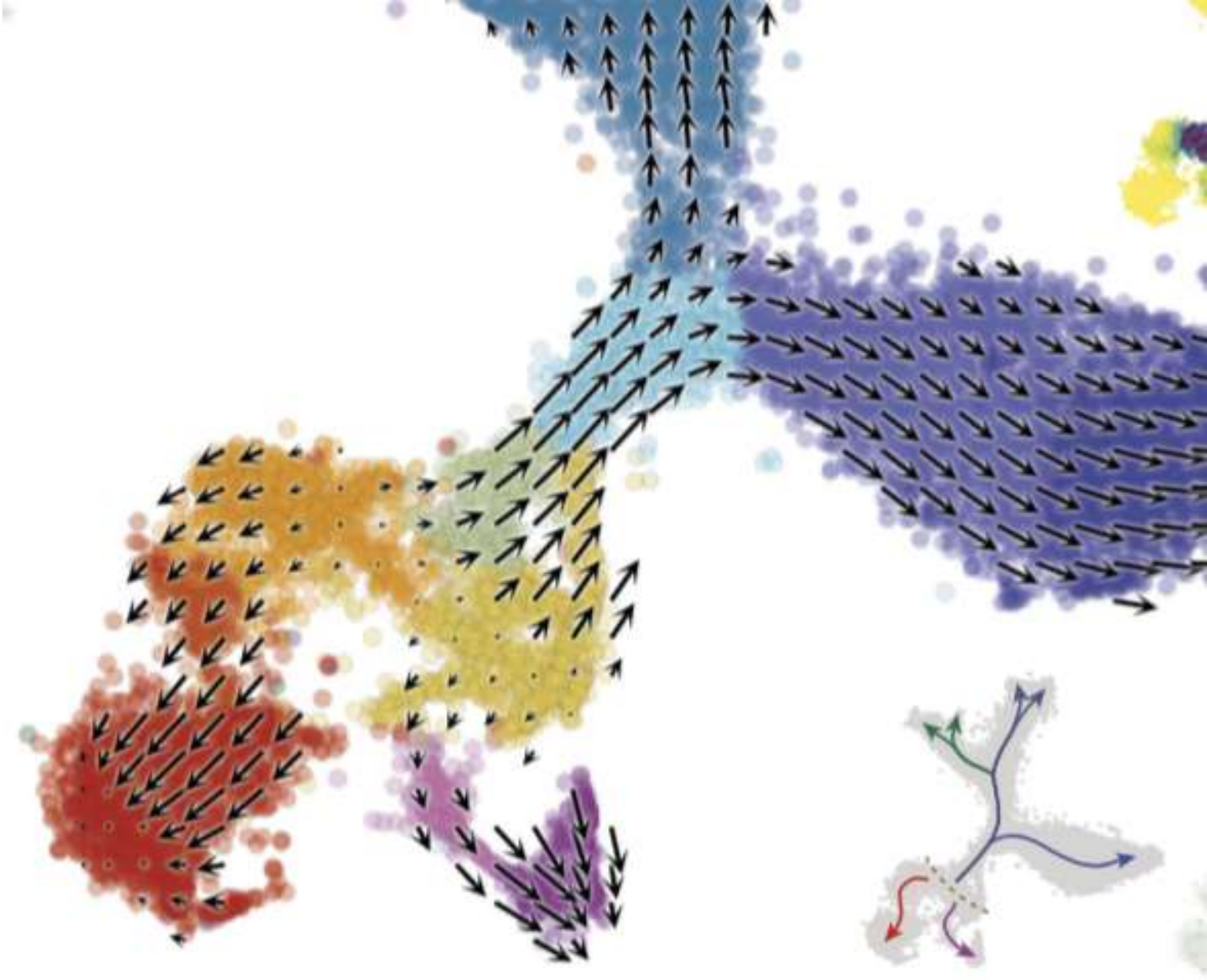


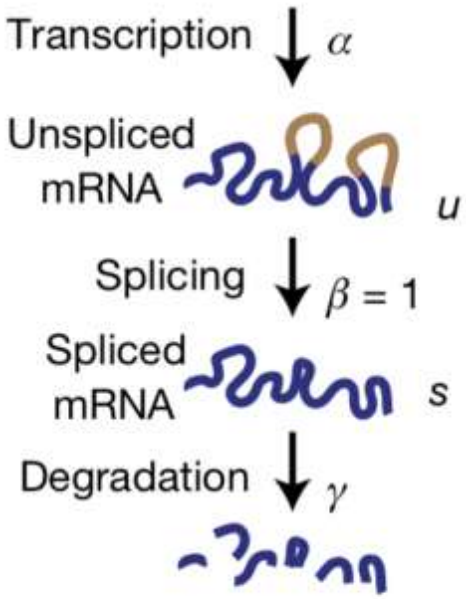




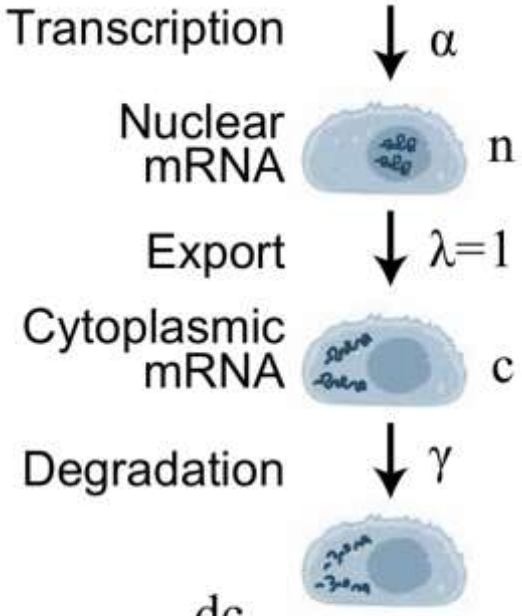

CA3
CA2/4
Subiculum
CA1
Granule
Neuroblast
Radial glia
nIPC
OPC
Astrocyte



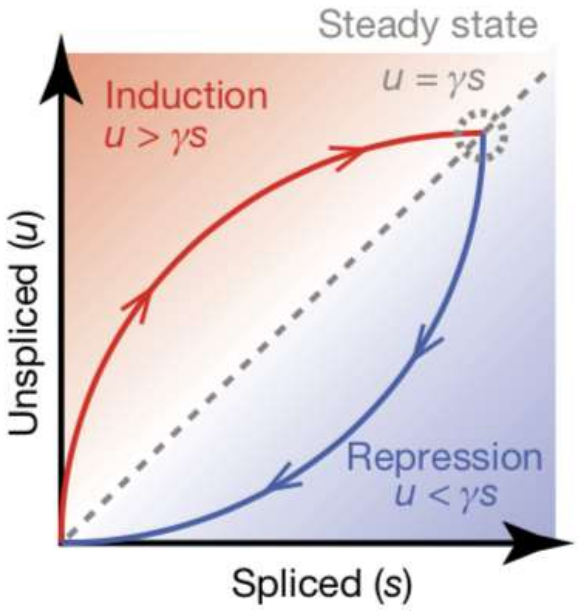
Adapt RNA velocity model for in situ analog using ratio of nuclear vs. cytoplasmic mRNAs

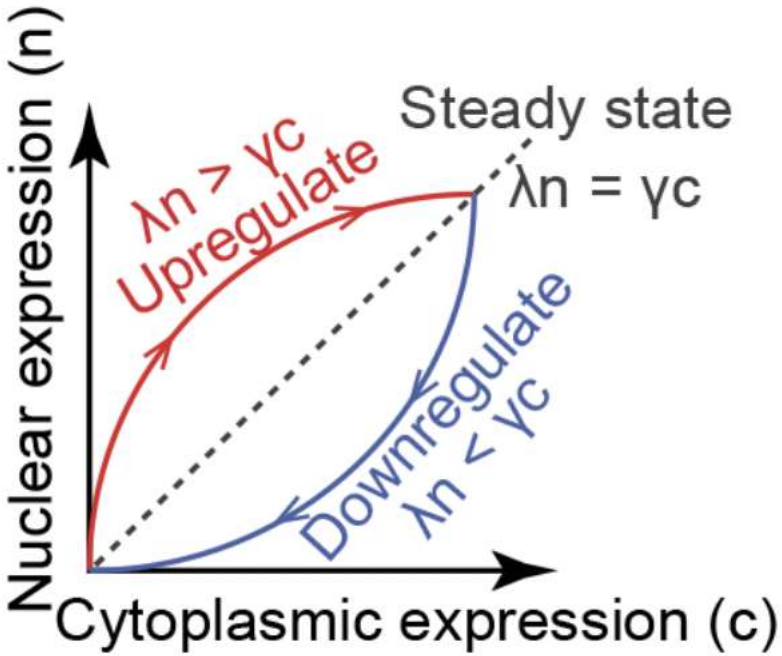


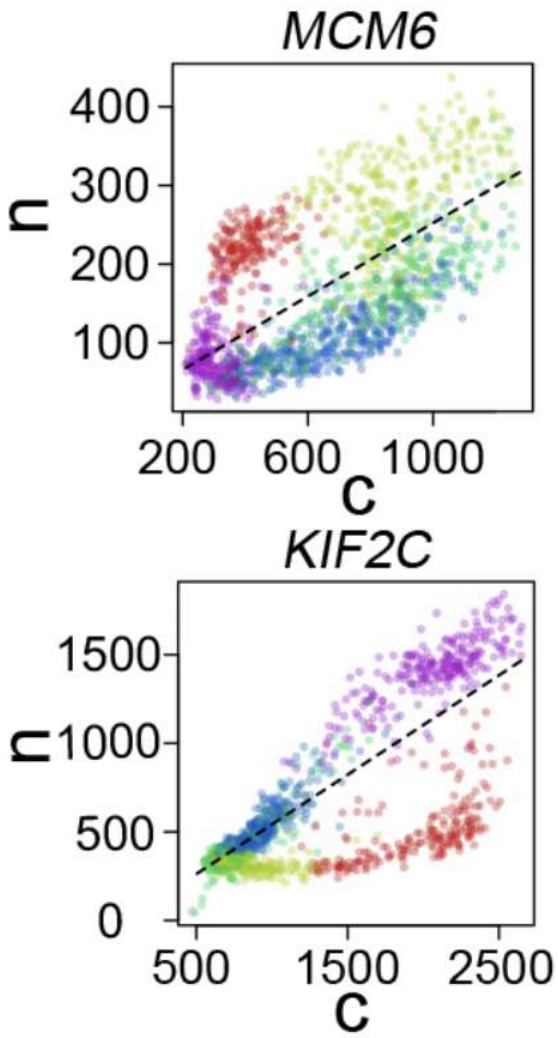
$$\frac{ds}{dt} = u - \gamma s$$

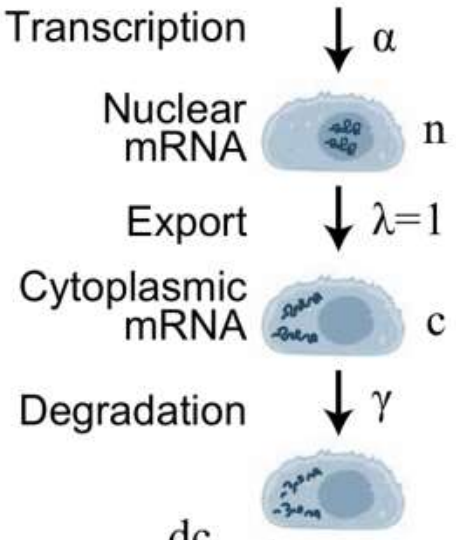


$$\frac{dc}{dt} = \lambda n - \gamma c$$

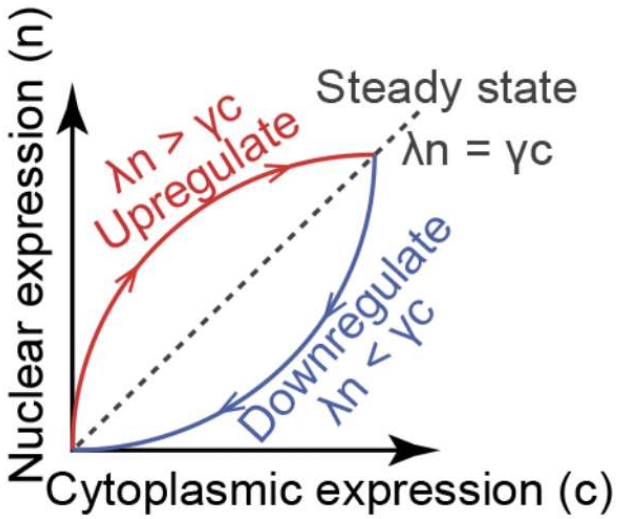

Steady state
$u = \gamma s$
Induction
$u > \gamma s$
Repression
$u < \gamma s$
Unspliced (u)
Spliced (s)


Nuclear expression (n)
Cytoplasmic expression (c)
Steady state
λn = γc
λn > γc
Upregulate
Downregulate
λn < γc


MCM6
400
300
200
100
n
200
600
1000
c
KIF2C
1500
1000
500
0
n
500
1500
2500
c



$$\frac{dc}{dt} = \lambda n - \gamma c$$



$$\frac{dn}{dt} = \alpha - n(t)$$

$$\frac{dc}{dt} = n(t) - \gamma * c(t)$$

At steady state:

$$\frac{dc}{dt} = n - \gamma c = 0$$

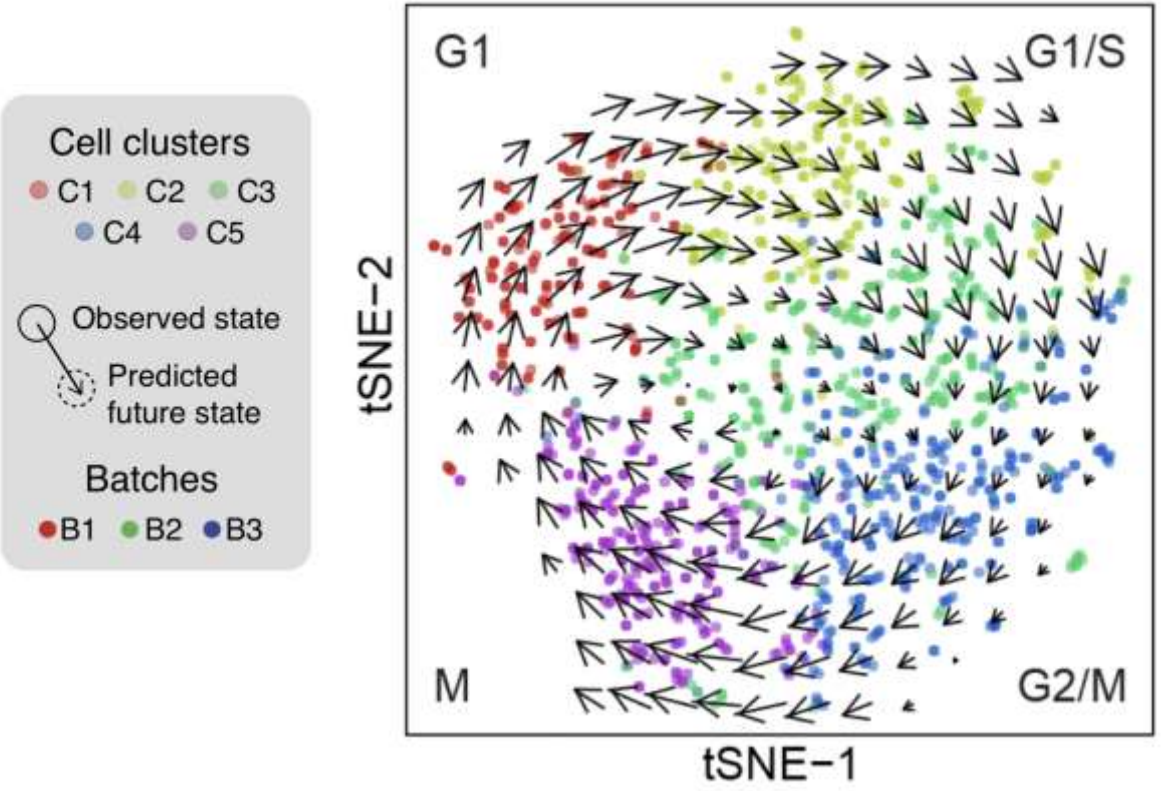
$$\gamma = \frac{n}{c}$$

Future transcriptional state:

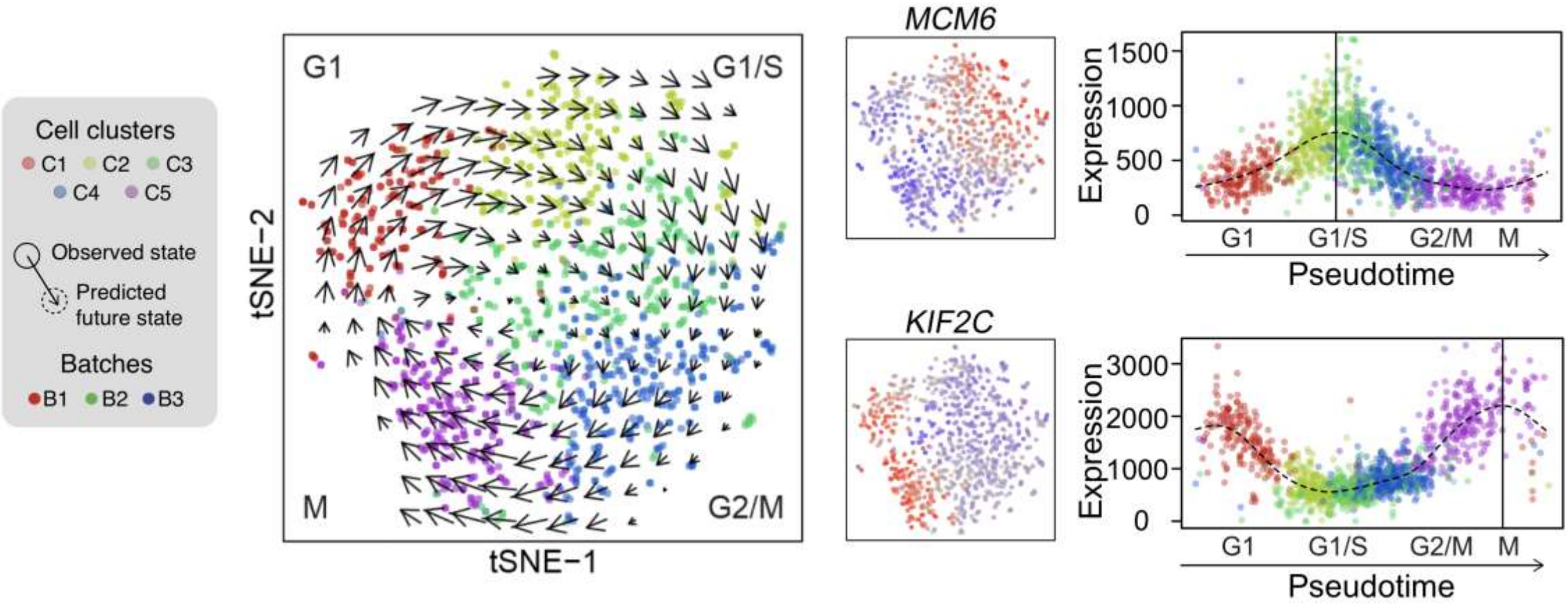
$$\frac{dc}{dt} = v = n - \gamma c$$

$$c(t + dt) = c(t) + vdt$$

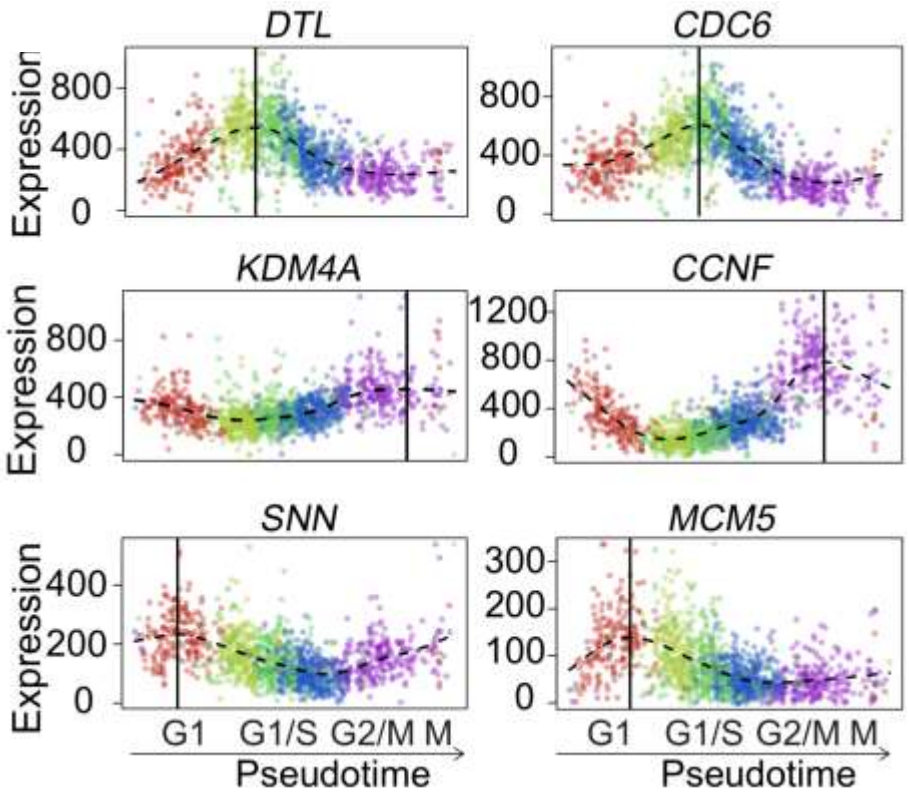
RNA velocity predicts future transcriptional state of cell to place cells in pseudotime

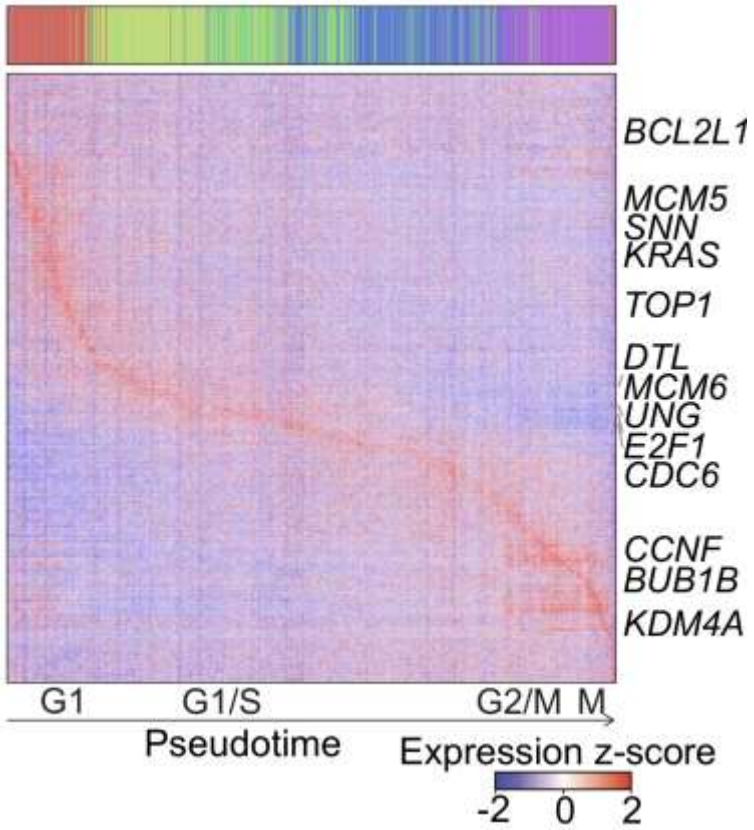


RNA velocity predicts future transcriptional state of cell to place cells in pseudotime



Pseudotime analysis identifies putative cell-cycle genes and highlights gradual transcriptional change throughout select cell-cycle stages





Summary

- MERFISH enables simultaneous spatial transcriptomic profiling for 10k genes with high efficiency and accuracy
- Combined with immunostaining identifies genes spatially co-localized at subcellular compartments
- Segmented into single-cells enables spatially-resolved transcriptomic profiling
- RNA velocity can be inferred in situ by discriminating nuclear and cytoplasmic mRNAs to predict future transcriptional cell state

Johns Hopkins University
July 2020
WE ARE HIRING!

contact jeanfan@fas.harvard.edu
for more info

Thanks and happy to take questions!

Zhuang Lab

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Ruslan Soldatov

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